

Internship Report On South Central Railway

**An
Internship
Report on
South Central Railway**

A report submitted in partial fulfillment of the requirements for the Award of Degree of

BACHELOR OF TECHNOLOGY

in

ELECTRICAL & ELECTRONICS ENGINEERING

by

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Regd. No:23K65A0224



Department of Electrical and Electronics Engineering

SASI INSTITUTE OF TECHNOLOGY&ENGINEERING

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada, and SBTET-Hyderabad,

Accredited by NAAC with 'A' Grade)

Kadakatla, Tadepalligudem– 532101

Academic Year 2025-2026

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CERTIFICATE

This is to certify that the internship work entitled “**South Central Railway**”(May-July) being submitted by **VENDRA LAKSHMI NARASIMHA (23K65A0224)** in partial fulfillment for the award of Degree of **BACHELOR OF TECHNOLOGY** in **ELECTRICAL & ELECTRONICS ENGINEERING** to the Jawaharlal Nehru Technological University, Kakinada during the academic year 2022-25 is a record of bonafide work carried out by them under our supervision.

Guide
Dr.D.Bala Bhaskar
Assistant Professor

Head of the Department
Dr. Avagaddi Prasad,
Prof & HoD

ACKNOWLEDGEMENT

I take immense pleasure to express my deep sense of gratitude to our beloved Guide **Dr.D.Bala Bhaskar**, Project guide for his valuable suggestions and rare insights, constant encouragement, and inspiration throughout the internship work.

I express my deep sense of gratitude to our beloved Principal, **Prof. Mohammed Ismail**, for his valuable guidance and for permitting us to carry out this internship.

I express my deep sense of gratitude to **Dr. Avagaddi Prasad**, Head of the Department for the valuable guidance and suggestions, keen interest shown and thorough encouragement extended throughout the period of internship work.

I am grateful to my internship coordinator and thanks to all teaching and nonteaching staff members who contributed for the successful completion of our internship work.

With Gratitude

V.Lakshmi Narasimha

23K65A0224

DECLARATION

The internship report entitled “***South Central Railway***” is a record of bonafide work carried out by me, in partial fulfillment for the award of degree of Bachelor of Technology from Jawaharlal Nehru Technology University Kakinada.

Submitted by

V.Lakshmi Narasimha

23K65A0224

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VISION & MISSION

Vision of the Institute:

- Promoting excellence in electrical and electronics engineering through knowledge impartation, skill cultivation, and graduate empowerment.

Mission of the Institute:

- To provide an ambience that is conducive for lifelong learning with an ability to solve real world challenges.
- To advance as a center of excellence with a thrust on industry research challenges.
- To cultivate in students a commitment to professionalism with utmost integrity and ethical standards.

Vision of the Program:

- Expand as a Centre of excellence in the field of Electrical Engineering through industrial and academic research by training the learners for global acceptance

Mission of the Program:

To work with commitment for the improvement of quality teaching, to conduct the creative research by addressing the needs of the industry and society and to develop the professional practice among the learners to encourage lifelong learning, teamwork and leadership.

PROGRAM EDUCATIONAL OBJECTIVES

Program Outcomes:

PO1: Engineering knowledge:

- Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2 – Problem Analysis:

- Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.

PO3 – Design/Development of Solutions:

- Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required.

PO4 – Conduct Investigations of Complex Problems:

- Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions.

PO5 – Engineering Tool Usage:

- Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems.

PO6 – The Engineer and The World:

- Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment.

PO7 – Ethics:

- Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws.

PO8 – Individual and Collaborative Team work:

- Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9 – Communication:

- Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO10 – Project Management and Finance:

- Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11 – Life-Long Learning:

Recognize the need for, and have the preparation and ability for

- independent and life-long learning

- adaptability to new and emerging technologies and
- critical thinking in the broadest context of technological change.

Program Educational Outcomes:

1. Apply the knowledge to design and optimize wired and wireless communication systems, embedded systems, signal and image processing, and Internet of Things (IoT) solutions.
2. Design and develop cost effective solutions to communication at higher frequency using modern simulators and VLSI development tools.

Program Specific Outcomes:

PSO 1: Able to apply knowledge and skill sets that solve problems and develop solutions to challenges in the field of power systems, power electronics, control systems and evolving technologies.

PSO 2: Able to adapt to the contemporary advancements in technology and provide low-cost solutions to the needs of society at large.

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**INTRODUCTION:
INDIAN RAILWAYS**

"Indian Railways.... the golden Era"

Railways were first introduced to India in 1853. By 1947, the year of [India's independence](#), there were forty-two rail systems. In 1951 the systems were nationalized as one unit, becoming one of the largest networks in the world. IR operates both long distance and suburban rail systems on a [multi-gauge](#) network of [broad](#), [meter](#) and [narrow gauges](#). It also owns [locomotive](#) and [coach](#) production facilities.

The introduction of Railways in our country has taken place at the juncture when the transportation system in our country was primitive with the introduction of Railways the transportation system and it changed a lot. It was in 16th April, 1953. When the first train was steamed of from Bombay to Thana. A stretch of 34 Kms. At present it is having a route length of 62.915 Kms. Indian Railways is not only the biggest public sector undertaking in the country but also the biggest utility concern. "Indian Railway" is largest in Asia and fourth in the world.

Since its introduction the "Indian Railway" is still continuing to be the predominant in Transportation. In the initial stages the administrative control of Railways in India was in the hands of the public works departments. But in 1903, the Robertson committee recommended for separate board for the Railways. This Board come into begin in February 1905 with authority to manage the Railways of India on the basis of Commercial principles.

The Board was initially in the hands of Ministry of Commerce and Industry but was late transferred to Ministry of Transport and communications. After Independence, the Railways are under the direct control of the Ministry of Railways assisted by the Railway Board in the matter of policy and decision making. For administrative convenience the Railways, Board re divided in 16 zones. Every zone is headed by a "General Manager" the zones the further divided in to different divisions and various departments for the better management.

The Indian Railways is the principal mode of transport in our country. In 150 years of its existences it has successfully adopted to the charging need of Transportation

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of goods and passengers in the country. Every year Indian Railway carries over 11 million passengers per day and lifted more than a million tones of freight traffic daily on network spread over 62,915 route Kms covering 7,068 stations.

The railways networks consists of Board gauge and Narrow gauges totaling a length of 1,08,336 track Kms. Gauge wise breakup on 31st March, 2006 and zones and their respective head quarters, jurisdiction of Divisions are shown in table 1 and table 2.

Table - 1

Particulars of gauge wise length of Railways lines in India.

In additional zonal railways cited above the following production units and projects are functioning directly under the control of Railway Board.

- Integral Coach Factory - Chennai
- Chittaranjan Loco Motive Works - Chittaranjan
- Diesel Locomotive Works - Varanasi
- Rail Wheel Factory - Bangalore
- Railway Coach Factory - Kapurthala
- Diesel Loco Modernization Works - Patiala

Rolling stock companies of Steam Engines diesel and electric & Diesel hydraulic Engines stationed at various loco sheds.

In Southern Railways, we have following the Diesel Shed & Electric Sheds.

Erode Diesel Sheds, Golden Rock (Work Shop), Krishnarajapuram (Bangalore)

Electrical Shed : Arakonam

South Central Railway : Kazipet, Guntakal, Moula-Ali

Electrical Shed : Vijayawada, Lalaguda

Manufacturing units for diesel locomotives at Varanasi (Diesel Locomotive Workshop) for Electrical (Chittaranjan Locomotives Workshop) and ICF for coaches (Integral Coach Factory), Kapurthala coaches manufacturing Railway Wheel factory at Bangalore.

RAILWAY ADMINISTRATION:

The Ministry of Railway functions under the guidance of the minister of Railways, assisted by the Minister of State for Railways. The day-to-day affairs and formulation of policy are managed by the Railway Board comprising of a Chairman, five other function members and a financial commissioner. The chairman of the Railway Board is the ex-officio Principal Secretary and each of the other five functional

members are the ex-officio Secretaries to the Government of India.

1.1 HISTORY AND ORGNISATION OF RAILWAYS:

INDIAN RAILWAYS: NATION'S LIFE LINE:

Indian Railways have continuously influence the social, economic, cultural and political life of the people. Some of the important aspects of this influence are increased mobility of the common man, bringing together people from all castes, communities and religions, growth trade and commerce development of industries, equalizing price line, urbanization, employment generation, rural development, help in spread of education, contribution to national exchequer, influence on literature, poetry, music files etc., Thus, Railways have become Nation's life line.

THE BIGGEST ORGANISATION:

Indian Railways are the biggest organization our country with 14.12 lakhs of employees, 63,332 route kilometers traversing from Okha to Murkong Selekand Udhampur to Kanyakumari, 6,974 Railway stations, a fletch of 8,025 locomotives, 50,080 passenger and other coaches. 2, 07,176 goods wagons, carrying 5,725 million originating passenger traffic and 682.4 million metric tones of originating goods traffic. If has a total investment of 1,12,180.41 crore rupees.

BRIEF HISTORY:

The first train in the world was run between Stockton and Darlington in England in 1825. In 1844, R.M. Stephenson submitte a propasal to East India Company to start the Railway system in India and it was accepted. The first contract was given to East Indian Railway Company in 1849. Later, great Indian peninsular Railway Company and other companies were given similar contracts in different parts of India. The first train ran on 16th April 1853 from Porbunder to Thane followed by the second from Howrah to Hoogly on 15th August 1854

The development in Railways in India has passed through various phases which are broadly as under:

1)	Guarantee System	-	1849 - 1869
2)	State Construction	-	1869 - 1882
3)	Modified Guarantee	-	1882 - 1924
4)	Nationalisation	-	1924 - 1948
5)	Integration and Regrouping	-	1948 - 1951

- 6) Era of Planning - 1951 onwards

RAILWAY'S MANAGEMENT:

Initially, the Board of Directors in England had an agent in India for supervision and control over the companies. The Government appointed advisory engineers and auditors. In 1855, Railways were placed under Public Works Department. A railway branch was organized under PWD in 1869 and a State Railway Directorate in 1874. Ultimately, the Railway Board was organized in 1905. The separate Ministry of Railways was formed after independence.

RAILWAY BOARD:

Railway Board is the apex executive body which administers, directs and supervised the functioning of the Railway system. The Board functions under the Minister for Railways and is headed by the Chairman, who is equivalent to Principal Secretary to Government of India. The other members are (1) Financial Commissioner, (2) Member Staff, (3) Member Traffic, (4) Member Mechanical, (5) Member Engineering and (6) Member Electrical. The Board has several directorates. Major directorates are headed by an Advisor. Health Services and RPF directorates have Director Generals. They are assisted by Executive Directors, Directors, Joint Directors etc.,

A. Board's Attached and Subordinate Offices etc.,

- (1) Research Design and Standards Organisation, Lucknow

Note: w.e.f.01.01.'03, RDSO has been declared to be a Zonal Railway with headquarters at Lucknow.

- (2) Railway Liaison Officer with the Directorate General of Supplies and Disposals.

B. Subordinate Offices:

(1) **Railway Recruitment Boards:** at Ahmedabad, Ajmer, Allahabad, Bangalore, Bhopal, Bhubaneswar, Mumbai, Calcutta, Chandigarh, Guwahati, Gorakhpur, Jammu & Kashmir, Chennai, Malda, Muzaffarpur, Patna, Ranchi, Secunderabad and Thiruvanthapuram.

(2) Centralized Training Institutions:

Railway Staff College, Vadodara, Indian Institute of Civil Engineering, Pune, Indian Railways Institute of Signal Engineering and Telecommunications,

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Secunderabad, Indian Railways Institute of Mechanical & Electrical Engineering, Jamalpur, Indian Railways Institute of Electrical Engineering, Nashik, Indian Railway Institute of Transport Management, Lucknow.

(3) Production Units:

- Chittaranjan Locomotive Works, Chittaranjan,
- Integral Coach Factory, Perambur, Chennai,
- Diesel Locomotive Works, Varanasi,
- Rail Wheel Plant, Bangalore,
- Diesel Component Works, Patiala,
- Railway Coach Factory, Kapurthala

Other Organizations:

- Metro Railways at Calcutta, Mumbai and Chennai
- Circular Railway Project, New Delhi
- Central Organisation for Modernization of Workshops, New Delhi,
- Central Organisation for Railway Electrification, Allahabad
- CRIS, New Delhi,
- Indian Railway Conference Association, New Delhi
- Railway Transport Museum, New Delhi

Commissions, Tribunals etc.,(Not under the Ministry) commissioner for Railway safety with offices at lucknow ,calcutta ,Bangalore and mumbai; Railway Rates Tribunal, chennai, Railway claims Tribunals, Railway Audit Organisation and other Commissions etc., organizes from time to time.

Undertakings under the Ministry:

- Rail India Technical and Economic Services Ltd., (RITES);
- Indian Railway Construction Company Ltd.,(IRCON);
- Indian Railway Financial Corporation;
- Indian Container Corporation Ltd.,
- Konkan Railway Corporation (CRC);
- Centre for Railway Information Systems (CRIS);
- Indian Railway Catering and Tourism Corporation Ltd.,(IRCTC);
- Rail Tel Corporation of India,
- Rail Tel Vikas Nigam Ltd.,(RVNL);

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- Pipavav Railway Corporation Ltd.,(PRCL)

1.2 ZONAL RAILWAYS:

The Indian Railways were divided into 09 zones.

Name of Railways	Formed on	HQ	Route Kms.
Southern Railway	14.04.1951	Chennai	5,210
Central Railway	05.11.1951	Mumbai CST	3,832
Western Railway	05.11.1951	Mumbai CG	6,533
Eastern Railway	14.04.1952	Kolkata	2,412
Northern Railway	14.04.1952	New Delhi	6,910
North Eastern Railway	14.04.1952	Gorakpur	3,402
South Eastern Railway	01.08.1955	Kolkata	2,614
North Eastern Frontier Railway	15.01.1958	Maligaon	3,766
South Central Railway	02.10.2002	Secunderabad	5,743
North Western Railway	01.10.2002	Jaipur	5,452
East Central Railway	01.10.2002	Hajipur	3,439
East Coast Railway	01.04.2003	Bhubaneswar	2,431
South Western Railway	01.04.2003	Hubli	3,111
West Central Railway	01.04.2003	Jabalpur	2,965
North Central Railway	01.04.2003	Allahabad	3,080
South East Central Railway	01.04.2003	Bilaspur	2,432

Each zonal railway is controlled by a General Manager. The General Manager is assisted by Principle HOD namely Addl.GM, Sr.Dy.GM, Financial Advisor and Chief Accounts Officer, Chief Engineer, Chief Mechanical Engineer, Chief Operating Manager, Chief Commercial Manager, Chief Electrical Engineer, Chief Signal & Telecommunications Engineer, Controller of Stores, Chief Personel Officer, Chief Medical Officer and Chief Security Commissioner.

SOUTH CENTRAL RAILWAYS

It was created on 2 October 1966 as the ninth zone of Indian Railways.^[1] The six divisions of this railway have a total 5,752 route kilometres of track. Current General Manager of S.C. Railway is Mr. G.N.ASTHANA, an officer of [IRSME](#). From the days

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of steam-hauled locomotives and wooden plank seats, South Central Railway has come a long way, modernizing its system with the state of the art high-powered Diesel and Electric Locomotives, high speed telescopic Passenger Coaches, and higher axle load wagons, higher capacity track in all important routes, LED based multiple aspect colour light signalling with panel interlocking & solid state interlocking, and digital microwave and optic fibre cable communication system, ISDN exchanges etc. Over the years, South Central Railway has attained sufficient transportation output with adequate infrastructure development and technological upgrading to serve the regions in its jurisdiction. Safe operation of trains, expansion of network, modern passenger amenities, punctuality of trains, courteous service and cleanliness in stations and trains remain always the thrust areas of this Railway. Being a service oriented organization, South Central Railway provided Computerized Passenger Reservation System at 85 Stations/locations covering 96% of the berths available. In the arena of information dissemination to the rail customers, it has provided "139" Interactive Voice Response System (IVRS) for Reservation and train enquiry, National Train Enquiry System (NTES) for real-time information on movement of trains, Passenger Operated Enquiry Terminals (POET) with information on availability of accommodation and confirmation and Close Circuit Television (CCTV) for real time reservation availability status at all important stations in its system.

For mass movement of freight, S.C. Railway has introduced high horse powered [diesel](#) and [electric locomotives](#) and high speed, higher axle load Box-N-Wagons.

In 2008 South Central Railways also had the privilege to be the first one to implement wireless MU Coupler system supplied by [Ms Lotus Wireless Technologies](#), permitting Longer freight train delivering higher capacity. The system was first implemented in Diesel Locomotive in 2008 followed by installation on Electric Locomotive in 2011 at Electric Loco Shed, Kazipet. The Trains equipped with these systems are in use at Badrachalam region in the South Central Railways. South Central Railway plays a pivotal role as a catalyst for agricultural and industrial development in the Southern peninsula apart from fostering the growth of trade and commerce including import/export through ports by connecting sea ports with their hinter land and inland container depots. Its reliable and comfortable passenger services for long and short distance travel by introducing many super fast and intercity trains helps transform the

society by catering to their personal, business, educational and tourism purposes. Jurisdictional adjustments were made in October 1977 by merging Guntakal Division of the Southern Railway with South Central Railway

and transferring Sholapur Division back to Central Railway. Secunderabad Division was split in February 1978 into two Divisions Secunderabad and Hyderabad to facilitate effective operational and administrative control. On 1 April 2003, the newly formed [Guntur division](#) and Nanded Division of South Central Railway became operational and Hugli Division was transferred to the newly formed [South Western Railway](#). Presently, the South Central Railway has six divisions, namely Secunderabad, Hyderabad, [Vijayawada](#), [Guntur](#), Guntakal and Nanded with 5752 Route km of which 1604 Rkm are [electrified](#). [Vijayawada](#) is the second largest railway junction in India and the most important station of the Indian Railway. Vijayawada is heart of south central railways it is more important than Secunderabad earning more income than rest of the other

South Central Railway provided Computerized Passenger Reservation System at 85 Stations/locations covering 96% of the berths available. In the arena of information dissemination to the rail customers, it has provided Inter-active Voice Response System (IVRS) for Reservation and train enquiry, National Train Enquiry System (NTES) for real time information on movement of trains, Passenger Operated Enquiry Terminals (POET) with information on availability of accommodation and confirmation and Close Circuit Television (CCTV) for real time reservation availability status at all important stations in its system.

SCR has made various advancements for reliable and comfortable Passenger Services for long and short distance travel by way introducing many super fast and intercity trains helps transform the society by catering to their personal, business, educational and tourism purposes.

2.1 SOUTH CENTRAL RAILWAY

Departments:

1. ETTC(Electric Traction Training Center)
2. RRI(Route Relay Interlocking)(Belongs to Signal)
3. OFC(Optical Fiber Communication)(Belongs to Telecommunication)

department)

4. IOH (Coach care center) and Engine depts (belongs to mechanical dept and electrical depts)

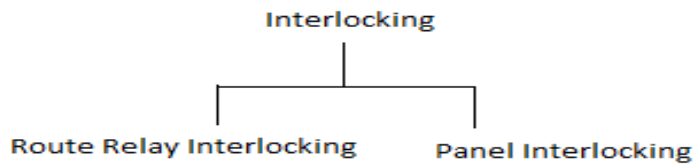
2.2 ROLE OF SIGNAL AND TELECOMMUNICATION DEPT.

- Safety in train operations
- Punctuality of running of trains
- Optimum utilization of line capacity
- Passenger amenities
- Manufacturing of specified signaling items
- Providing telecommunication infrastructure for :
 - a. Safe and efficient operation of trains
 - b. Administrative voice and data circuits
 - c. Passenger Reservation System (PRS)
 - d. Unreserved Ticketing System (UTS)
 - e. Freight Operation Information System (FOIS)
 - f. Crew Management System (CMS)
 - g. Coaching Operating Information System (COIS)
 - h. Management Information System (MIS)
- Management of Railways Corporate Wide Information System (RAILNET)

3. ELECTRIC TRACTION TRAINING CENTRE

3.1 INTERLOCKING

Interlocking is a mechanism to check the route the train is going to traverse before clearing the signal and to prevent the clearing of signal when it is unsafe for the movement. It eliminates human error and ensures safety in train operations.



3.2 PANEL INTERLOCKING

Panel interlocking is provided for smaller yards. It is a signaling system, which assures the best standard of safety and eliminates the accidents arising out of human errors.

4. ROUTE RELAY INTERLOCKING (RRI)

4.1 Introduction

Route Relay Interlocking was first introduced in India at Basin Bridge Junction and Madras Central Stations of Southern Railway.

This was followed by the installations in the suburban stations like in Bangalore.

This system of RRI is popular in Southern Railway and South Central Railway.

RRI also exists in South Eastern and North Eastern Railways.

Route Relay Interlocking is provided at major junctions and yards where large numbers of train movements are required. It is available at four major junctions/yards viz. Vijayawada, Kazipet, Secunderabad & Dornakal stations in South Central Railway.

4.2 Description of System

This system is based on a geographical layout of signal switches and push buttons. Operator should know how to move trains like where a particular train is and to which line it has to be routed. Therefore, to set up a particular route, the operator has to perform some action on the geographical layout of Control Panel at these two points at *Entrance* and *Exit*. In British System, these two actions take the form of turning a switch at the entrance and pressing a button at the Exit, of that particular route. For this reason, the system is known as "Entrance-Exit System" or "NX System".

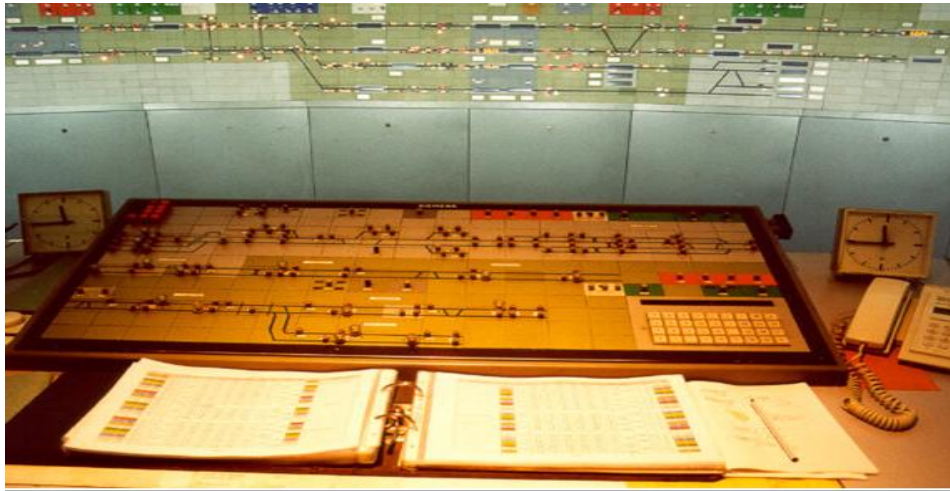


Fig1:Railway Signal Control Panel
Control Panel



Fig2 : Centralized Traffic Control Panel in Railway Signaling System

All the signals and points are operated from a cum control panel located in the Station Master's Room.

Control Panel consists of an inclined console on which a clear geographical representation of track layout with signals, points, Control switches, pushbuttons and

various indications are available. The track layout is sub-divided into track sections. There are different colors for each track circuit section (except Red). On the track, adjacent to each signal there is an "Entrance Thumb Switch" at the center of the track of the route, and an "Exit push Button". Exit button is colored White, Entrance switch is colored Red for running signals and Yellow for shunt (to change tracks) signals with the number of the relevant signals engraved in the center of the switch. The king knob used in semi-automatic territory is colored red with a white bar across the knob. There are two types of signal switches, one turning 90° towards right and the other turning 90° towards left. Normally, the switch knobs will be at right angles to the track and when turned they become horizontal, the knob pointing towards the direction of the train movement that has to take place. Push button restores to normal when released.

In addition to Signal switches, 3-Position point thumb switches colored black are provided, one for each set of points for individual operation of points. Calling ON Signal switch is colored Red with white dot and crank handle releasing switch is blue in color. Similarly level crossing control switch is colored in deep brown. When a route is set and locked, the route is illuminated by white strip lights in the track circuit configurations throughout the route (except the overlap). This indication remains lit as long as the route is locked and disappears only after the relevant signal switch is restored to normal position and the route is released. This indication turns to Red when track is occupied or track circuit failed, irrespective of whether the route is locked or free. When the speed and frequency of the trains are to be increased, Relay interlocking is ideal. In Relay Interlocking the Interlocking is achieved through relay circuitry at a centralized place. The time required for installations is also less. The Relay Systems are generally trouble free since the moving parts involved are less and replacement is easier. Indian Railways are having time tested experience in Relay Interlocking since 1958. Route Relay Interlocking is a system in which Interlocking is achieved by means of Relays. In British system the relays conforming to BRS and BSS specification, having metal to carbon contacts are used, (Q Series Relays). Therefore, the circuitry becomes comparatively simpler as welding of contacts need not be considered.

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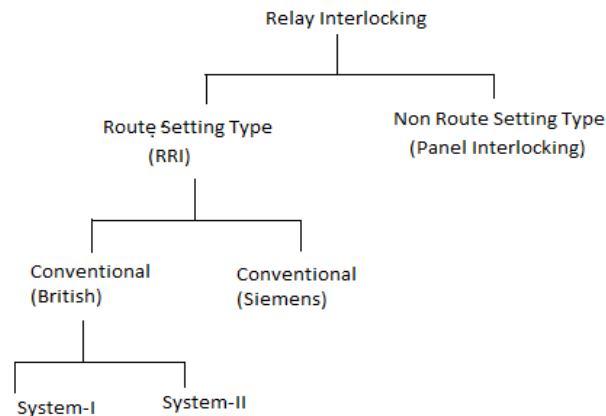


Fig 3: Types of Relay Interlocking in Railway System

OPERATIONS:

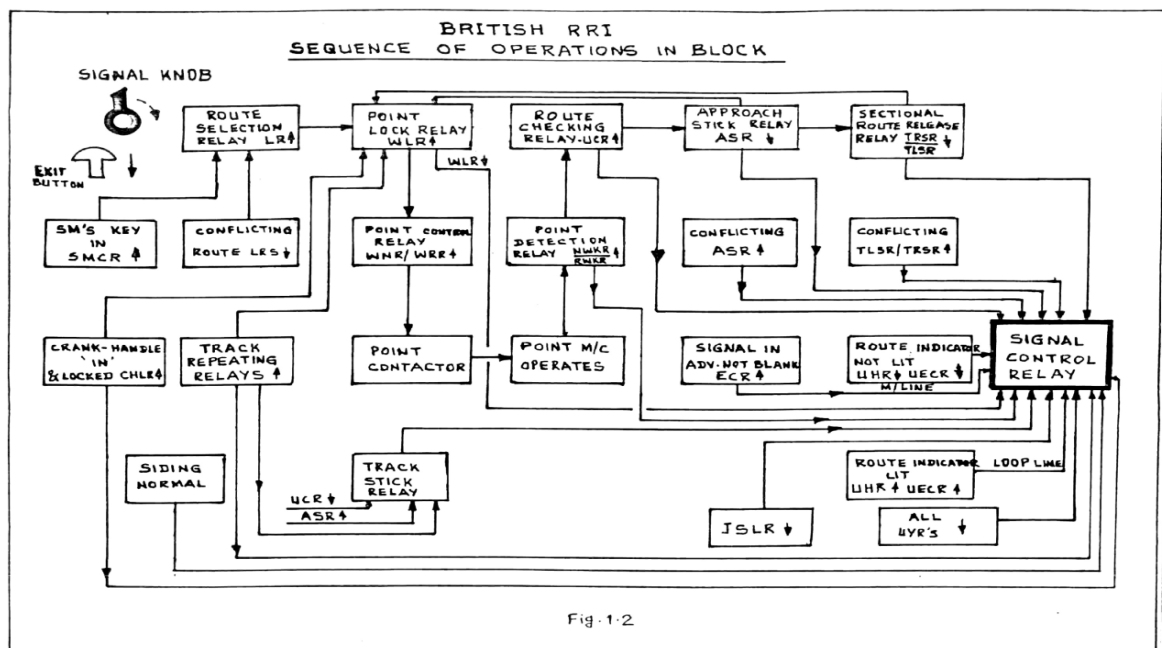


Fig4 : Block Diagram of British Railway Route Signaling Operation

Turning the entrance knob and pressing the exit button of a route, energizes the route selection relay, provided that no conflicting route is set. Thus the basic interlocking is ensured at the first stage itself.

The energisation of the route selection relay picks up all the point control relays in that route depending on the route selected provided that the points are free from track locking and route locking. The point control relay controls the point machine concerned and sets the points required in the route. The correct setting and locking of each point is indicated by the point indication relay.

The route checking relay checks that all the points involved in the selected route are correctly set and locked at the site. It also proves that the route set is for the Signal route initiated including isolation and overlap.

The operation of the route checking relay de-energizes the relevant approach stick relay and sectional route locking relays thereby ensuring that the complete route is locked before the signal is cleared.

The signal Control Relay is energized proving all safety conditions required viz., all tracks, including overlap are clear, all points, including those in overlap, isolation, are correctly set and locked, relevant route locking and approach locking relays are de-energized etc. In addition a control by track stick relay is used to ensure that the signal does not re-clear after the passage of a train as the relevant thumb switch is left in the operated position. The control by the track stick relay is removed if the signal has to work as an automatic signal, by the operation of king knob relay, which is energized by the reversal of the king knob.

The approach stick and sectional route locking relays will re-energize when the train arrives & clears the route and the signal switch is turned to Normal Position. Sectional release of route is, of course, provided wherever necessary.

In British System, after the arrival of a train on proper signals, route gets released only after the normalization of concerned signal knob by the operator provided all other conditions are satisfied. Since the Route is released with the knowledge of Station Master, the overlap will be released immediately, as soon as the Route is released. In British System holding of overlap for certain time delay (120 minutes) is not considered mandatory.

5. OPTICAL FIBRE COMMUNICATION

5.1 TELECOMMUNICATION

Providing basic telecommunications facilities for the Railways vast transport system is the responsibility of the Signal & Telecom Department. The telecommunication facilities provided on the Railways are broadly categorized in to three areas of applications.

-Train Operation

-Data Communication for Railway Information Technology Applications

-Administrative Communications.

TELECOMMUNICATION FOR TRAIN OPERATION

Provides omnibus communication circuits on OFC and Quad cable and overhead telecom wires for operation of trains and other supporting activities of train control. The omnibus circuit provides telephone communication between the station-masters and the control center at the divisional headquarters. Other omnibus circuits are also provided for other supporting train operations, including remote operation of 25 KV AC power supply systems required for electric traction. Emergency communication facility is provided along with the track throughout the route for the drivers and guards of the running train to communicate to the control center at the respective divisional headquarters during emergencies.

Optical Fiber Communication system is the backbone of Railways Telecommunication Network. The short-haul STM1 equipment of 155 Mbps capacity is provided at every station along the fiber network. The short-haul STM1 network is protected by long-haul STM4 / STM16 self-healing ring network provided by RailTel. The fiber network carries Voice, Data & Multimedia services for various applications.

OFC PROVIDES FOLLOWING TYPES OF COMMUNICATION NEEDS

- Control Communication with emergency communication
- Administrative voice and data communication.
- Both for control and long haul communication backbone for mobile train radio communication.

GUIDELINES FOR USE OF OPTICAL FIBRE SYSTEM ON VARIOUS ROUTES

On A, B, C, D & D Spl Routes, following Optical Fiber system

Configuration should be used.

- a) 24 Fibers Optical Fiber Cable with Six Quad cable Short Haul Communication will be on STM1 System and Primary Digital Multiplexers (PDMUX). For back up a second pair of Fiber shall be used.
- b) On other routes, following Optical Fiber System configuration should be used: Composite Cables (8/24 Fiber + 6 Quad) shall be used. Short Haul Communication will be on STM1 System and Primary Digital Multiplexers (PDMUX). For back up a second pair of Fiber shall be used.
- c) Replacement of existing overhead/RE quad cable for control Communication.

CAPACITY OF FIBRE OPTIC SYSTEM

- a. For Control Communication application, SDH System having capacity of STM1 shall be used. Generally, one E1 shall be used for various control applications.
- b. For Long Haul Communication, SDH System with capacity of STM4 or above shall be used.

OPTICAL COMMUNICATION BLOCK DIAGRAM

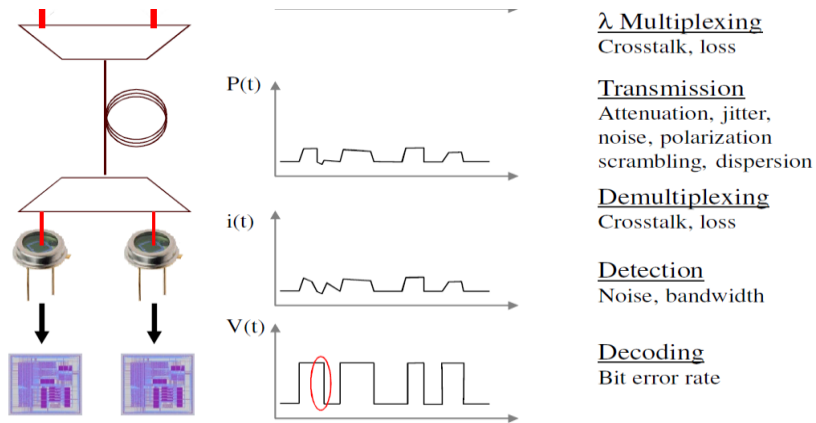


Fig 5: Block Diagram of WDM Transmission System

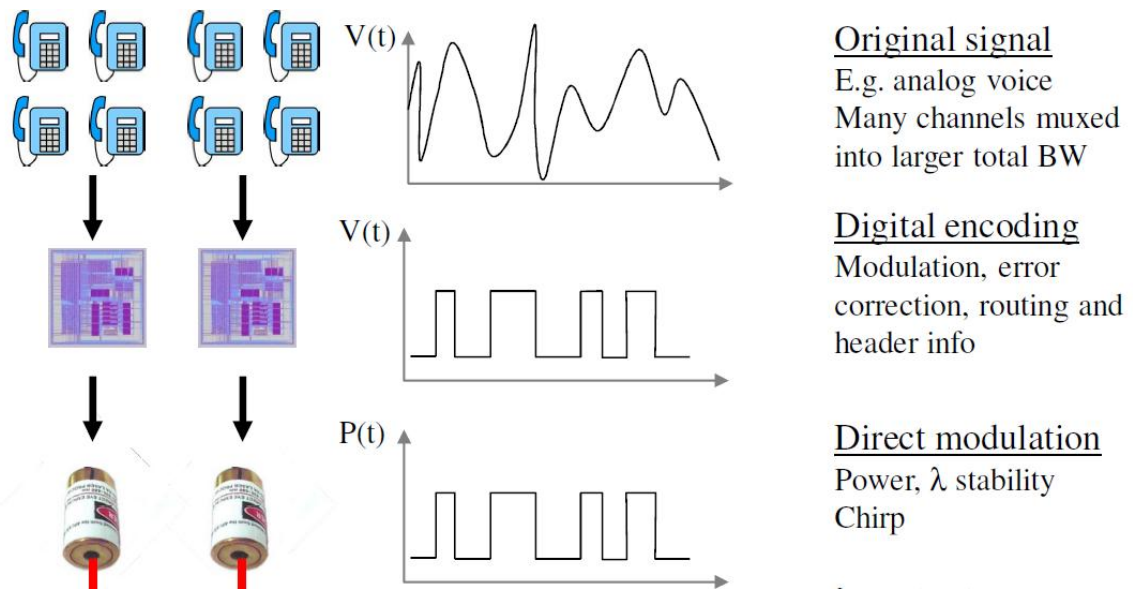


Fig6 : Conversion of Analog Voice Signals into Digitally Modulator Transmission

Time-division multiplexing

Bundling low bandwidths

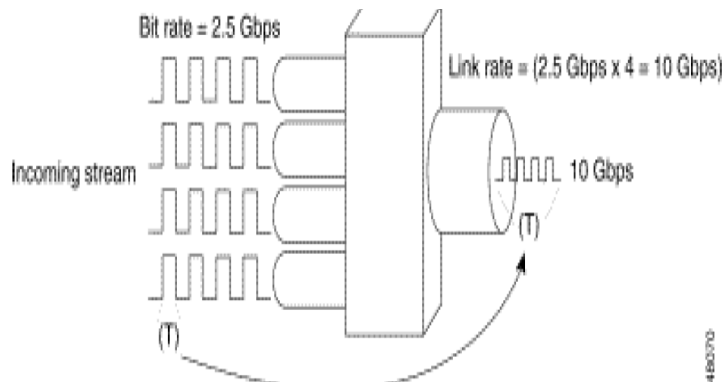


Fig7 : Time-division multiplexing Process

- Electronic aggregation of multiple low-bandwidth streams.
- Eventually exceed available modulation rate of electronics and optics.
- Standard today is 10 Gb/s with 40 Gb/s being fielded.
- Optical carrier (at $\lambda = 1.5 \mu\text{m}$) is 200 THz, so there is room for a number of these TDM channels
- Available bandwidth is determined by fiber

Wavelength division multiplexing to exploit transmission bandwidth

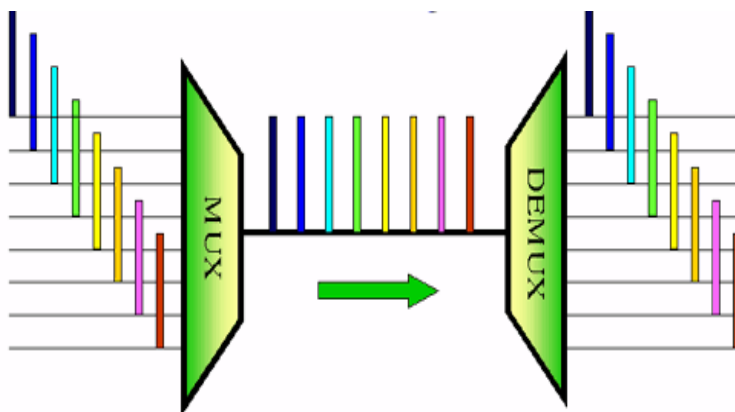


Fig 8: Process of Multiplexing and Demultiplexing in WDM

- Also known as frequency multiplexing

- Mux/demuxes need flat pass bands in both amplitude and phase to not distort the data, then very sharp filter edges to suppress crosstalk between channels.
- Gratings are an obvious technology but are difficult to package with sufficient stability.
- Daisy chains of thin film band-drop filters can be used
- Array waveguide gratings – essentially an integrated optic grating –are becoming the dominant technology

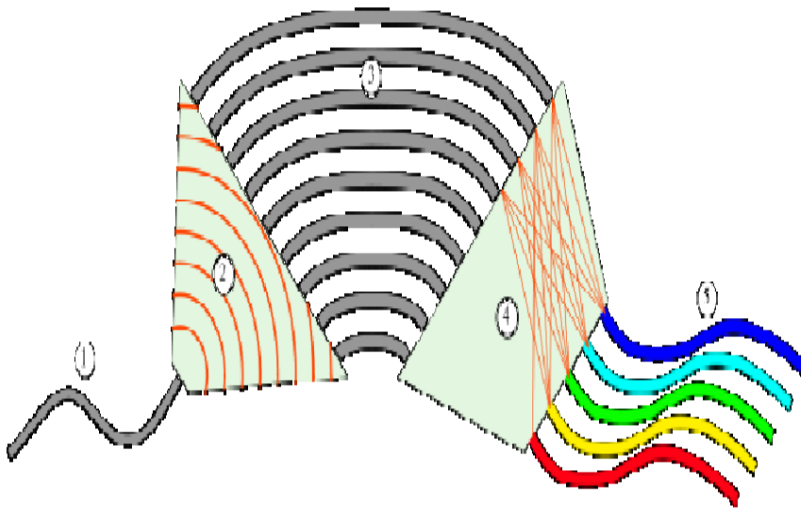


Fig9 : Wave Length Separation Using Prism in WDM System

Fiber optics primer

Capture via total internal reflection

$$\theta_c = \sin^{-1}(n_1/n_2) \quad (1-2)$$

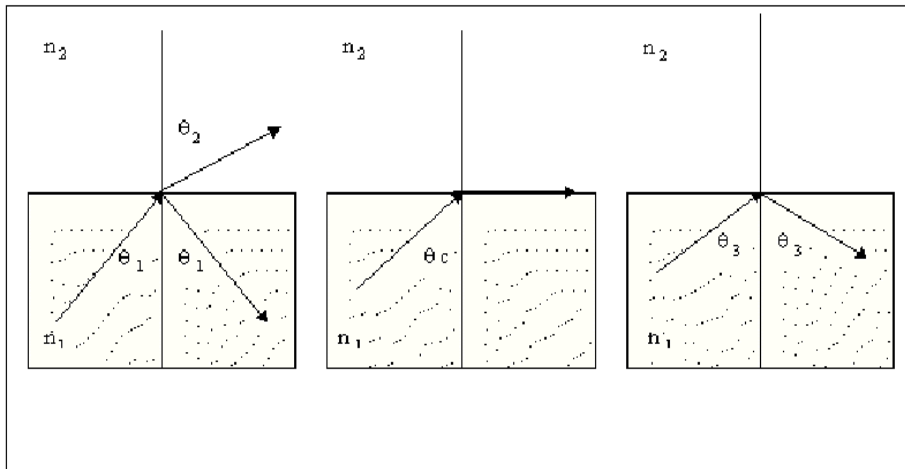


Fig10 : Light Propagation Through Different Refractive Indices

- For a perfectly flat surface, TIR is a 100% efficient mirror
- Once captured inside a higher index region, light will never exit
- Bending the guide decreases the incidence angle resulting in optical loss

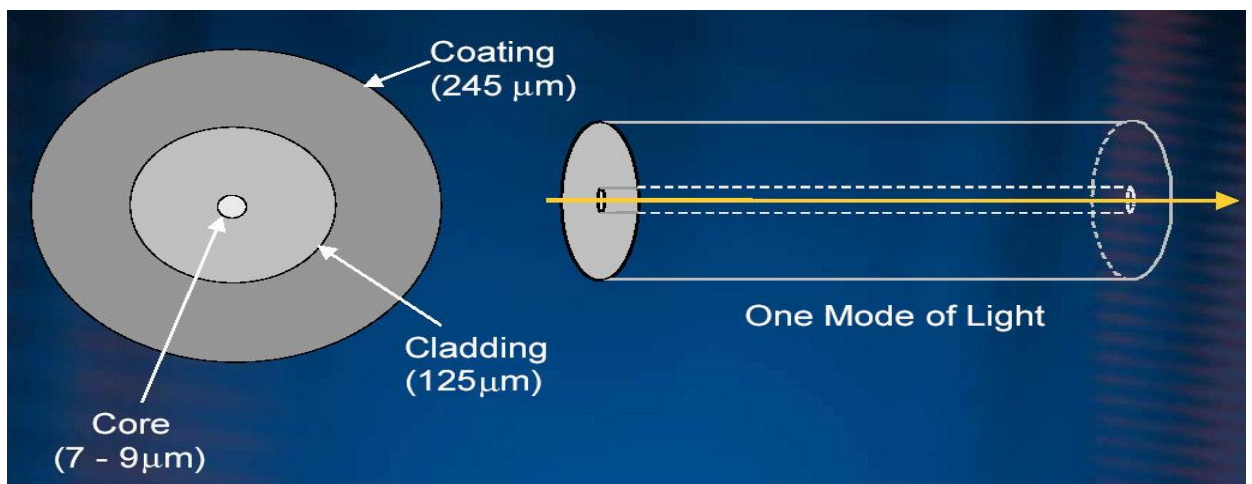
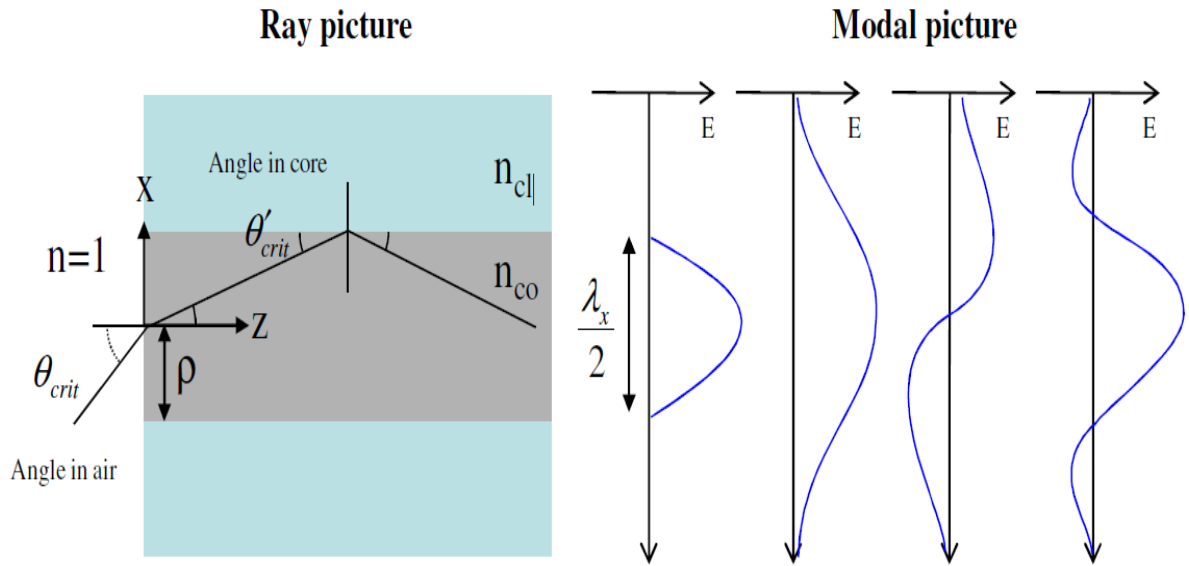


Fig11 : Cross Section and Mode Structure Of a Single Mode Fiber

Waveguide basics

Important normalized quantities



Ray view of guiding in a slab waveguide. The most extreme ray is trapped via total internal reflection at the core/cladding boundary.

Mode 0 of a metal guide.

Modes 0,1,2 of a dielectric guide.

Fig12 : Ray and Model Representations Of Light Propagation in Waveguide

Numerical aperture of guide measured in air

$$NA \equiv \sin \theta_{crit} = \sqrt{n_{co}^2 - n_{cl}^2}$$

Phase shift of limiting wave across guide face

$$V \equiv k_x \rho = k_0 \sin \theta_{crit} \rho = k_0 NA \rho = k_0 \rho \sqrt{n_{co}^2 - n_{cl}^2}$$

Interpretation of $V = 2 \pi \rho / \lambda_x = \pi D / \lambda_x = \pi (\# \text{ waves across face})$.
So $V = \pi/2$ is cutoff for metal waveguide, bit larger for index guides

Fiber drawing

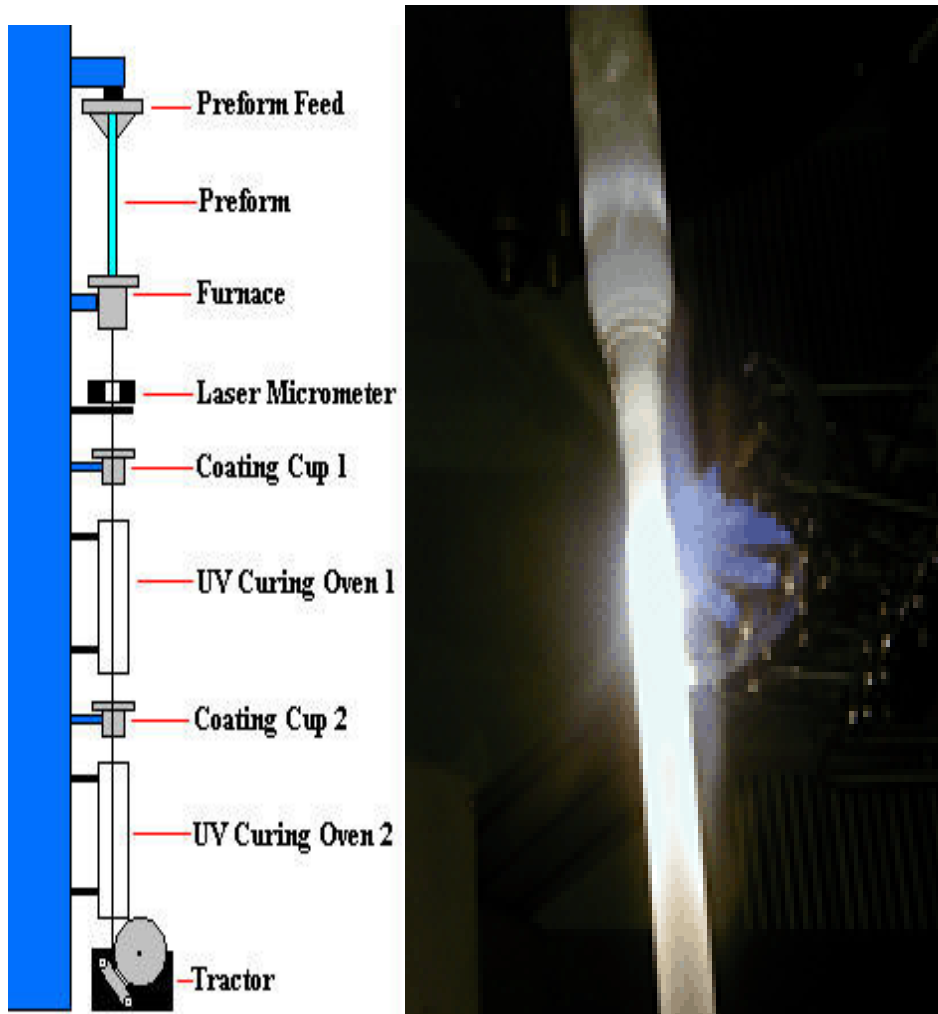


Fig13 : Optical Fiber Drawing Process and Manufacturing Setup

5.2 FIBRE MODES

Single Mode System

Source = laser diode. Aperture size and NA related by diffraction limit.(Numerical Aperture)

Fiber = single mode Mode diameter and NA related by diffraction limit.

Can be nominally 100% efficient.

Coupled must be precisely aligned on fiber axis. Tight tolerances (~micron or less)

High brightness system.

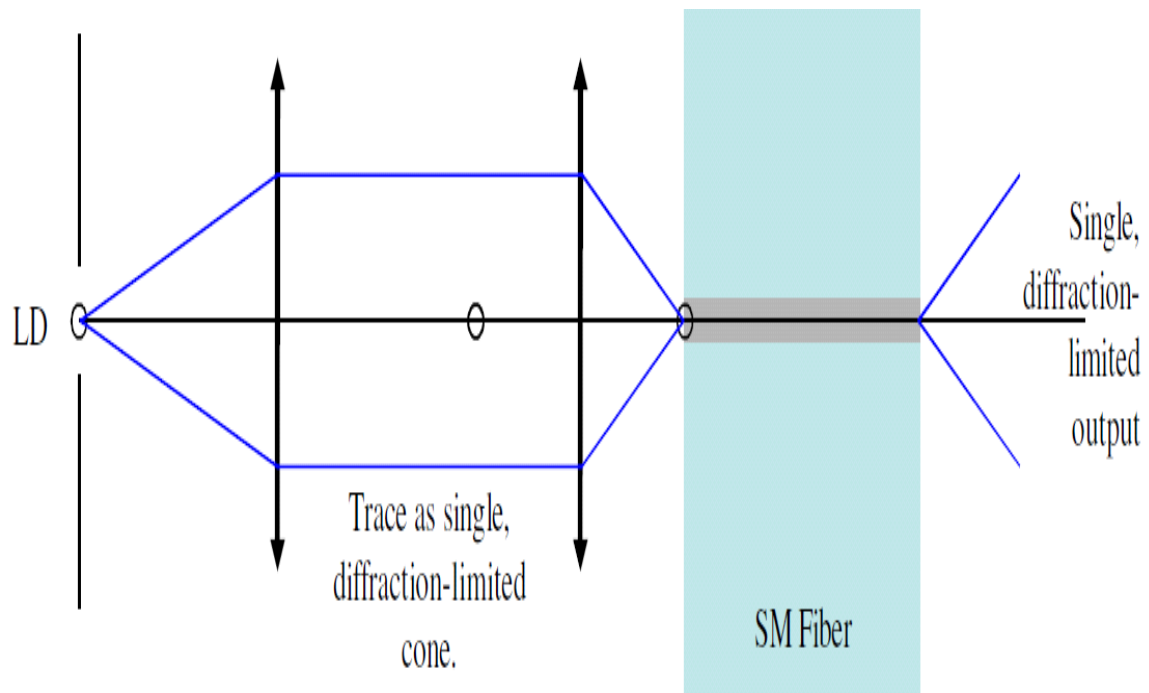


Fig14 : Coupling of Laser Diode Light Into a Single Mode Optical Fiber

Multi-Mode System

Source = light-emitting diode. NA ~ 1 independent of aperture diameter

Fiber = multi-mode Finite NA, unrelated to core diameter

Large losses in first lens due to LED radiation angle

Any ray that hits core with $\sin(\text{angle}) < \text{NA}$ is coupled. Loose tolerances ($\sim 10 \mu\text{m}$).

Lower brightness system

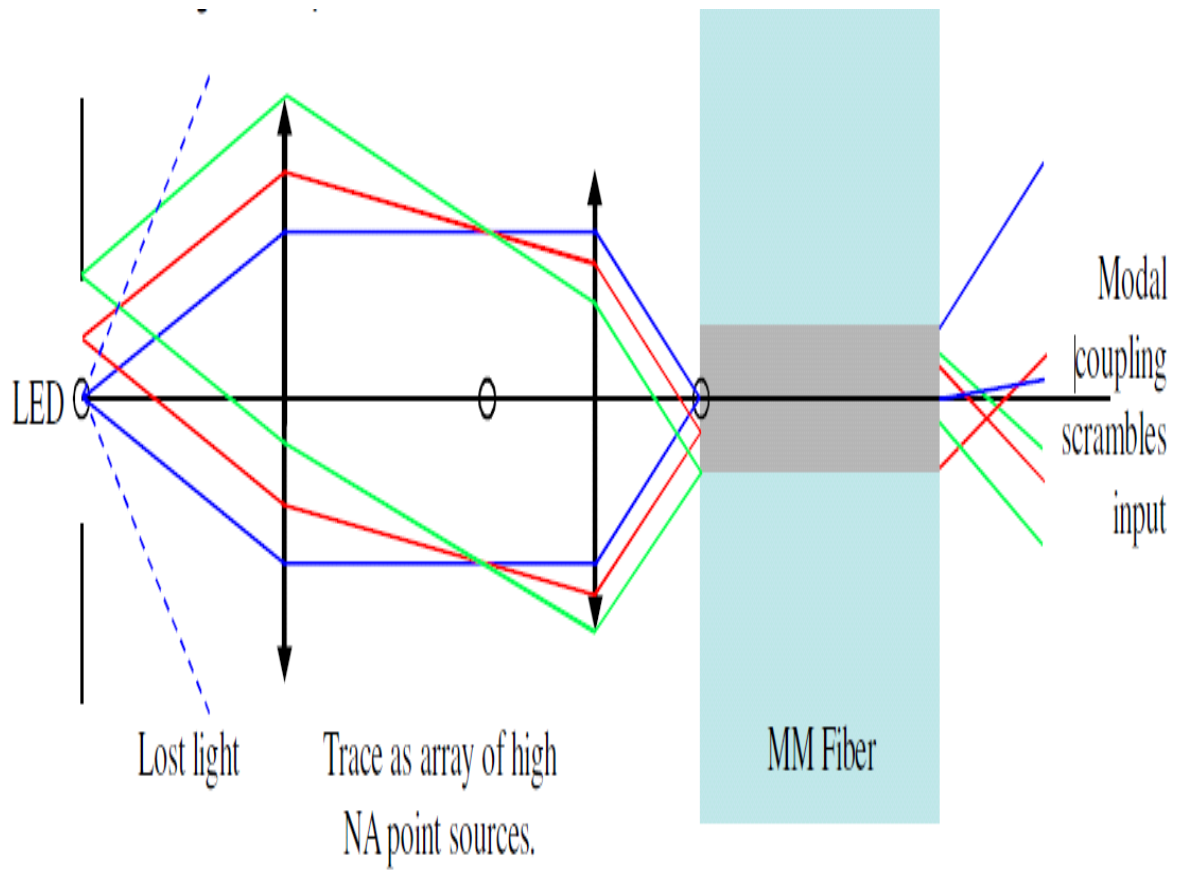


Fig15 : Light Propagation and Mode Coupling in Multimode Fiber



Fig16 : Signal Communication

Single Mode Vs Multi Mode Fibers

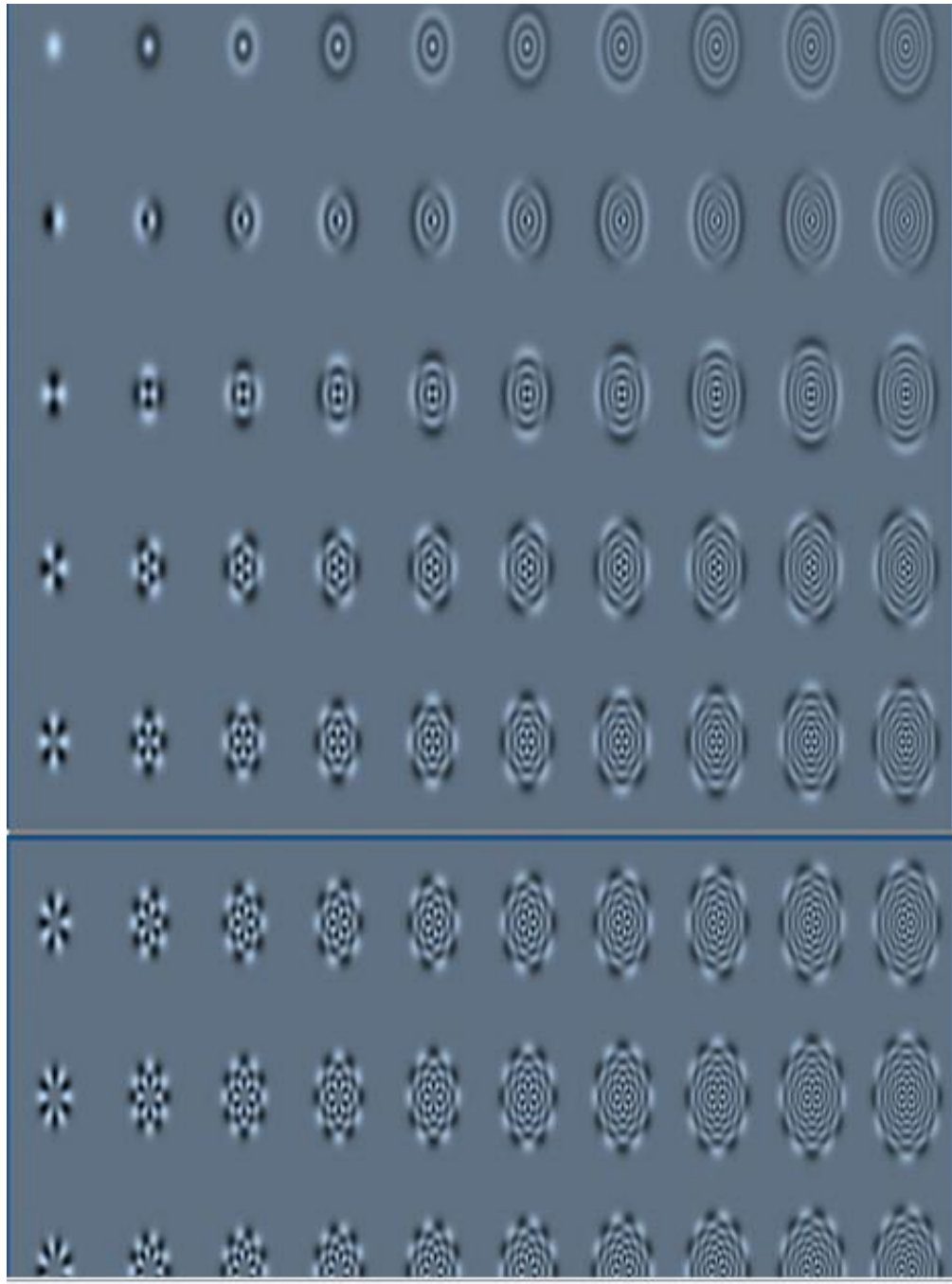


Fig17 : Transverse Mode Patterns in a Multimode Optical Fiber

6 ENGINE

1.The activities of Diesel Loco Shed, Vijayawada:

Integrated Maintenance of Diesel Locomotives & Diesel Electrical Multiple Units (DEMUs), Rail Buses, Self-Propelled Accident Relief Train & 140Ton Break Down Crane to cater in Vijayawada, Guntur divisions.

- a. DME
- b. ADME (Diesel)
- c. ADME (DEMU)
- d. AMM (Diesel)

2.ADMINISTRATION:

The shed is headed by One DME assisted by 02 ADMEs and 01 AMM. The shed is having 350 Man power (includes Technical Supervisors, Artisan staff for loco maintenance, Ancillary Staff, Ministerial staff, 44 members towards maintenance of Break Down crane & SPART and other miscellaneous staff). One Store department attached to DLS/BZA is headed by one AMM with 1012 Nos. stocked items.

3.BRIEF EXPLANATION:

6.1 DIESEL LOCO SHED (DLS)

ENGINE:

Engine is a device which converts the chemical energy into mechanical energy.

Types of engines based on the fuel used:

- 1. Petrol engine
- 2. Diesel engine
- 1. Petrol engine:

A Petrol engine is an engine which uses fuel as petrol.

It is a spark ignition engine in which spark is initiated by spark plug.

- 2. Diesel engine:

The essential part of the diesel locomotive is engine which uses diesel as fuel.

This is a compression ignition engine.

The fuel in the diesel engine's cylinders is ignited by the heat caused by air being suddenly

compressed in the cylinder.

The compression ratio of 25 to 1.

Advantages of diesel engine over petrol engine are:

1. High thermal efficiency.
2. Fuel is cheaper.
3. It can do heavy work under over loads.
4. No need of spark plug to initiate the spark.
5. More torque by only few gear changing.

Disadvantages of diesel engine over petrol engine:

1. More noisy in high speed condition.
2. More expensive.
3. Parts of diesel engine are very expensive.
4. Diesel vehicles tend to emit more particulates than petrol.

*In railways diesel engines are used.

Types of diesel engines based on number of strokes:

There are two types of diesel engines which are:

1. Two stroke engine
2. Four stroke engine

Two stroke engine Four stroke engine

It has no valves and has ports It has two valves.

It has 2 strokes completed in 1 revolution. It has 4 strokes complete in 2 revolutions.

Downstroke:suction,compression

Upstroke: power, exhaust.

Downstroke-1:suction

Upstroke-1:compression

Downstroke-2:power

Upstroke-2:exhaust

Functioning of Four strokes:

Four strokes will take place those are SUCTION, COMPRESSION, POWER, and EXHAUST.

In the Suction stroke inlet valves are opened and the air will be taken into cylinder by moving the piston from top dead centre (TDC) to bottom dead centre (BDC) and then in

compression stroke the piston moves from BDC to TDC and air will be compressed and at the end of compression stroke fuel will be sprayed in atomized form which gently mix with air and will burn in power stroke from TDC to BDC after that exhaust stroke will be from BDC to TDC at end of this stroke the exhaust gases will be sent out by opening the exhaust valve.(OR) In suction stroke Air is drawn into the cylinders. In Compression stroke the piston squeezes,

or compresses, the air up to the top. Compressing the air makes it very hot and at the same

instant, a valve sprays diesel fuel into the top of the cylinder. Heat from the compressed air sets fire to the fuel. In Power stroke the exploding fuel makes a lot of pressure, which

pushes the piston down. The up and down movement of the pistons turns the crankshaft. In Exhaust stroke a valve at the top lets out the waste gas, and the cylinder takes in more air for the next stroke.

Functioning of two strokes engine:

In two stroke engine downward stroke engine comprises of both suction and compression.

upward stroke comprises of power and exhaust strokes.

*In railways mainly 4 – stroke engines are used.

Breif explanation about diesel locomotive:

6.1 DIESEL LOCOMOTIVE:

The loco is generally divided into following divisions:

Short Hood		Long Hood			
Nose Compartment	Driver Cab	Alternator Room	Engine Room	Expresser Room	Radiator Room

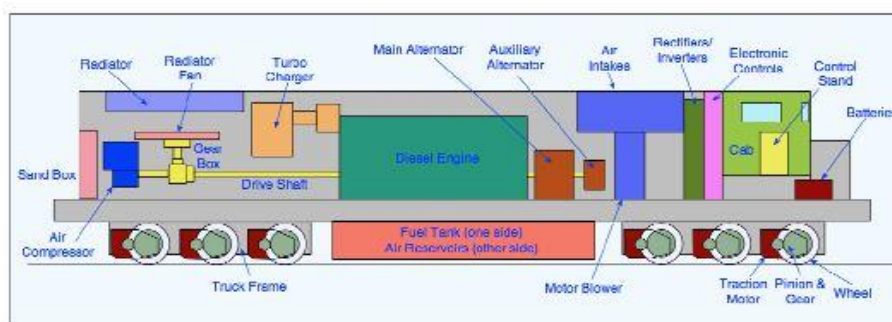


Fig18 : Layout and Division of a Diesel Locomotive

6.2 The Diesel Locomotive:

- The modern diesel locomotive is a self contained version of the electric locomotive.
- Like the electric locomotive, it has electric drive, in the form of traction motors driving the axles and controlled with electronic controls.
- It also has many of the same auxiliary systems for cooling, lighting, heating, braking and hotel power (if required) for the train.
- It can operate over the same routes (usually) and can be operated by the same drivers.
- It differs principally in that it carries its own generating station around with it, instead of being connected to a remote generating station through overhead wires or a third rail.
- The generating station consists of a large diesel engine coupled to an alternator producing the necessary electricity.
- A fuel tank is also essential.

Diesel Engine:

- This is the main power source for the locomotive.
- It comprises a large cylinder block, with cylinders arranged in a straight line or in a V.
- The engine rotates the drive shaft at up to 1,000 rpm and this drives the various items needed to power the locomotive.

Main Alternator:

- The diesel engine drives the main alternator which provides the power to move the train.
- The alternator generates AC electricity which is used to provide power for the traction motors mounted on the trucks (bogies).
- In older locomotives, the alternator was a DC machine, called a generator.
- It produced direct current which was used to provide power for DC traction motors.
- Many of these machines are still in regular use.
- The next development was the replacement of the generator by the alternator but still using DC traction motors.
- The AC output is rectified to give the DC required for the motors.

Auxiliary Alternator:

- Locomotives used to operate passenger trains are equipped with an auxiliary alternator.
- This provides AC power for lighting, heating, air conditioning, dining facilities etc. on the train.
- The output is transmitted along the train through an auxiliary power line. It is known as "head end power" or "hotel power".
- Air conditioned passenger coaches' get what is called electric train supply (ETS) from the auxiliary alternator.

Motor Blower:

- The diesel engine also drives a motor blower.
- As its name suggests, the motor blower provides air which is blown over the traction motors to keep them cool during periods of heavy work.
- The blower is mounted inside the locomotive body but the motors are on the trucks, so the blower output is connected to each of the motors through flexible ducting.
- The blower output also cools the alternators.
- Some designs have separate blowers for the group of motors on each truck and others for the alternators.
- Whatever the arrangement, a modern locomotive has a complex air management system which monitors the temperature of the various rotating machines in the locomotive and adjusts the flow of air accordingly.

Air Intakes:

- The air for cooling the locomotive's motors is drawn in from outside the locomotive.
- It has to be filtered to remove dust and other impurities and its flow regulated by temperature, both inside and outside the locomotive.
- The air management system has to take account of the wide range of temperatures from the possible +40°C of summer to the possible -40°C of winter.

Rectifiers/Inverters:

- The output from the main alternator is AC but it can be used in a locomotive with either DC or AC traction motors.
- DC motors were the traditional type used for many years but, in the last 10 years, AC motors have become standard for new locomotives.
- They are cheaper to build and cost less to maintain and, with electronic management can be very finely controlled.

- To convert the AC output from the main alternator to DC, rectifiers are required.
- If the motors are DC, the output from the rectifiers is used directly.
- If the motors are AC, the DC output from the rectifiers is converted to 3-phase AC for the traction motors.

Electronic Controls:

- Almost every part of the modern locomotive's equipment has some form of electronic control.
- These are usually collected in a control cubicle near the cab for easy access.
- The controls will usually include a maintenance management system of some sort which can be used to download data to a portable or hand-held computer.

Control Stand:

- This is the principal man-machine interface, known as a control desk or control stand.
- The common of stand is positioned at an angle on the left side of the driving position and, it is said, is much preferred by drivers to the modern desk type of control layout.

Cab:

- The standard configuration of locomotives is to have a cab at one end of the locomotive only.
- Since most structure gauge is large enough to allow the locomotive to have a walkway on either side, there is enough visibility for the locomotive to be worked in reverse.
- However, it is normal for the locomotive to operate with the cab forwards.
- In many countries, locomotives are full width to the structure gauge and cabs are therefore provided at both ends.

Batteries:

Just like an automobile, the diesel engine needs a battery to start it and to provide electrical power for lights and controls when the engine is switched off and the alternator is not running.

Traction Motor:

- Since the diesel-electric locomotive uses electric transmission, traction motors are provided on the axles to give the final drive.
- These motors were traditionally DC but the development of modern power and control electronics has led to the introduction of 3-phase AC motors.
- There are between four and six motors on most diesel-electric locomotives.

- A modern AC motor with air blowing can provide up to 1,000 hp. Pinion/Gear - The traction motor drives the axle through a reduction gear.

Fuel Tank:

- A diesel locomotive has to carry its own fuel around with it and there has to be enough for a reasonable length of trip.
- The fuel tank is normally under the loco frame and will have a capacity of say 1,000 imperial gallons.
- In addition to fuel, the locomotive will carry around, typically about 300 US gallons of cooling water and 250 gallons of lubricating oil for the diesel engine.

Air Reservoirs:

- Air reservoirs containing compressed air at high pressure are required for the train braking and some other systems on the locomotive.
- These are often mounted next to the fuel tank under the floor of the locomotive.

Air Compressor:

The air compressor is required to provide a constant supply of compressed air for the locomotive and train brakes.

Drive Shaft:

The main output from the diesel engine is transmitted by the drive shaft to the alternators at one end and the radiator fans and compressor at the other end.

Gear Box:

- The radiator and its cooling fan is often located in the roof of the locomotive.
- Drive to the fan is therefore through a gearbox to change the direction of the drive upwards.

Radiator and Radiator Fan:

- The radiator works the same way as in an automobile.
- Water is distributed around the engine block to keep the temperature within the most efficient range for the engine.
- The water is cooled by passing it through a radiator blown by a fan driven by the diesel engine.

Turbo Charging:

- The amount of power obtained from a cylinder in a diesel engine depends on how much fuel can be burnt in it.

- The amount of fuel which can be burnt depends on the amount of air available in the cylinder.
- So, if you can get more air into the cylinder, more fuel will be burnt and you will get more power out of your ignition.
- Turbo charging is used to increase the amount of air pushed into each cylinder.
- The turbocharger is driven by exhaust gas from the engine.
- This gas drives a fan which, in turn, drives a small compressor which pushes the additional air into the cylinder.
- Turbo charging gives a 50% increase in engine power.
- The main advantage of the turbocharger is that it gives more power with no increase in fuel costs because it uses exhaust gas as drive power.
- It does need additional maintenance, however, so there are some types of lower power locomotives which are built without it.

Sand Box:

- Locomotives always carry sand to assist adhesion in bad rail conditions.
- Sand is not often provided on multiple unit trains because the adhesion requirements are lower and there are normally more driven axles.

Truck Frame:

This is the part (called the bogie) carrying the wheels and traction motors of the locomotive.

Wheel:

- Wheels are driven by the power from the engine block.
- The wheels are connected to a traction motor by bull gears that transmit power to the wheels.

Mechanical Transmission:

- A diesel-mechanical locomotive is the simplest type of diesel locomotive.
- As the name suggests, a mechanical transmission on a diesel locomotive consists a direct mechanical link between the diesel engine and the wheels.

Fluid Coupling:

- In a diesel-mechanical transmission, the main drive shaft is coupled to the engine by a fluid coupling.
- This is a hydraulic clutch, consisting of a case filled with oil, a rotating disc with curved blades driven by the engine and another connected to the road wheels.

- As the engine turns the fan, the oil is driven by one disc towards the other.
- This turns under the force of the oil and thus turns the drive shaft. Of course, the start up is gradual until the fan speed is almost matched by the blades.
- The whole system acts like an automatic clutch to allow a graduated start for the locomotive.

Gearbox:

- This does the same job as that on an automobile.
- It varies the gear ratio between the engine and the road wheels so that the appropriate level of power can be applied to the wheels.
- Gear change is manual.
- There is no need for a separate clutch because the functions of a clutch are already provided in the fluid coupling.

Final Drive:

- The diesel-mechanical locomotive uses a final drive similar to that of a steam engine.
- The wheels are coupled to each other to provide more adhesion.
- The output from the 4-speed gearbox is coupled to a final drive and reversing gearbox which is provided with a transverse drive shaft and balance weights.
- This is connected to the driving wheels by connecting rods.

Hydraulic Transmission:

- Hydraulic transmission works on the same principal as the fluid coupling but it allows a wider range of "slip" between the engine and wheels. It is known as a "torque converter".
- When the train speed has increased sufficiently to match the engine speed, the fluid is drained out of the torque converter so that the engine is virtually coupled directly to the locomotive wheels.
- It is virtually direct because the coupling is usually a fluid coupling, to give some "slip".
- Higher speed locomotives use two or three torque converters in a sequence similar to gear changing in a mechanical transmission and some have used a combination of torque converters and gears.

Diesel Multiple Units (DMUs)

- The diesel engines used in DMUs work on exactly the same principles as those used in locomotives, except that the transmission is normally mechanical with some form of gear change system.
 - DMU engines are smaller and several are used on a train, depending on the configuration.
 - The diesel engine is often mounted under the car floor and on its side because of the restricted space available.
 - Vibration being transmitted into the passenger saloon has always been a problem but some of the newer designs are very good in this respect.
 - There are some diesel-electric DMUs around and these normally have a separate engine compartment containing the engine and the generator or alternator.
 - The reason for using one type or the other is really a question of preference.
- However, it can be said that the 2-stroke design is simpler than the 4-stroke but the 4-stroke engine is more fuel efficient.

Diesel-Electric Types:

Diesel-electric locomotives come in three varieties, according to the period in which they were designed.

These three are:

- DC - DC (DC generator supplying DC traction motors); AC - DC (AC alternator output rectified to supply DC motors) and AC-DC-AC(AC alternator output rectified to DC and then inverted to 3-phase AC for the traction motors).
- The DC - DC type has a generator supplying the DC traction motors through a resistance control system, the AC - DC type has an alternator producing AC current which is rectified to DC and then supplied to the DC traction motors and, finally, the most modern has the AC alternator output being rectified to DC and then converted to AC (3-phase) so that it can power the 3-phase AC traction motors.
- Although this last system might seem the most complex, the gains from using AC motors far outweigh the apparent complexity of the system.
- There is one traction alternator (or generator) per diesel engine in a locomotive.

Transmissions:

- Like an automobile, a diesel locomotive cannot start itself directly from a stand.

- It will not develop maximum power at idling speed, so it needs some form of transmission system to multiply torque when starting.
- It will also be necessary to vary the power applied according to the train weight or the line gradient.
- There are three methods of doing this: mechanical, hydraulic or electric.
- Most diesel locomotives use electric transmission called "diesel-electric" locomotive.
- Mechanical and hydraulic transmissions are still used but are more common on multiple unit trains or lighter locomotives.

Governor

- Once a diesel engine is running, the engine speed is monitored and controlled through a governor.
- The governor ensures that the engine speed stays high enough to idle at the right speed and the engine speed will not raise too high when full power is demanded.
- The governor consists of a rotating shaft, which is driven by the diesel engine.
- A pair of flyweights is linked to the shaft and they rotate as it rotates.
- The centrifugal force caused by the rotation causes the weights to be thrown outwards as the speed of the shaft rises.
- If the speed falls the weights move inwards.
- The flyweights are linked to a collar fitted around the shaft by a pair of arms.
- As the weights move out, so the collar rises on the shaft.
- If the weights move inwards, the collar moves down the shaft.
- The movement of the collar is used to operate the fuel rack lever controlling the amount of fuel supplied to the engine by the injectors.

Fuel Injection

- Ignition in a diesel engine is achieved by compressing air inside a cylinder until it gets very hot (say 400°C, almost 800°F) and then injecting a fine spray of fuel oil to cause a miniature explosion.
- The explosion forces down the piston in the cylinder and this turns the crankshaft.
- To get the fine spray needed for successful ignition the fuel has to be pumped into the cylinder at high pressure.
- The fuel pump is operated by a cam driven off the engine.
- The fuel is pumped into an injector, which gives the fine spray of fuel required in the cylinder for combustion.

Starting

- A diesel engine is started (like an automobile) by turning over the crankshaft until the cylinders "fire" or begin combustion.
- The starting can be done electrically or pneumatically.
- Pneumatic starting was used for some engines.
- Compressed air was pumped into the cylinders of the engine until it gained sufficient speed to allow ignition, then fuel was applied to fire the engine.
- The compressed air was supplied by a small auxiliary engine or by high pressure air cylinders carried by the locomotive.
- Electric starting is now standard.
- It works the same way as for an automobile, with batteries providing the power to turn a starter motor which turns over the main engine.
- In older locomotives fitted with DC generators instead of AC alternators, the generator was used as a starter motor by applying battery power to it.

Engine Control Development

- The systems used by most locomotives in service today are more sophisticated.
- To begin with, the drivers control was combined with the governor and hydraulic control was introduced.
- One type of governor uses oil to control the fuel racks hydraulically and another uses the fuel oil pumped by a gear pump driven by the engine.
- Some governors are also linked to the turbo charging system to ensure that fuel does not increase before enough turbocharged air is available.
- In the most modern systems, the governor is electronic and is part of a complete engine management system.

Tractive Effort, Pull and Power

- The definition of tractive effort (TE) is simply the force exerted at the wheel rim of the locomotive and is usually expressed in pounds (lbs) or kilo Newtons (kN).
- By the time the tractive effort is transmitted to the coupling between the locomotive and the train, the drawbar pull, as it is called will have reduced because of the friction of the mechanical parts of the drive and some wind resistance.
- Power is expressed as horsepower (hp) or kilo Watts (kW) and is actually a rate of doing work.

- A unit of horsepower is defined as the work involved by a horse lifting 33,000 lbs one foot in one minute.
- In the metric system it is calculated as the power (Watts) needed when one Newton of force is moved one metre in one second.
- The relationship between power and drawbar pull is that a low speed and a high drawbar pull can produce the same power as high speed and low drawbar pull.
- If you need to increase higher tractive effort and high speed, you need to increase the power.
- To get the variations needed by a locomotive to operate on the railway, you need to have a suitable means of transmission between the diesel engine and the wheels.
- One thing worth remembering is that the power produced by the diesel engine is not all available for traction.
- In a 2,580 hp diesel electric locomotive, some 450 hp is lost to on-board equipment like blowers, radiator fans, air compressors and "hotel power" for the train.

Power Control

- The diesel engine in a diesel-electric locomotive provides the drive for the main alternator which, in turn, provides the power required for the traction motors.
- We can see from this therefore, that the power required from the diesel engine is related to the power required by the motors.
- So, if we want more power from the motors, we must get more current from the alternator so the engine needs to run faster to generate it.
- Therefore, to get the optimum performance from the locomotive, we must link the control of the diesel engine to the power demands being made on the alternator.
- In the days of generators, a complex electro-mechanical system was developed to achieve the feedback required to regulate engine speed according to generator demand.
- The core of the system was a load regulator, basically a variable resistor which was used to vary the excitation of the generator so that its output matched engine speed.

The control sequence (simplified) was as follows:

- Driver moves the power controller to the full power position.
- An air operated piston actuated by the controller moves a lever, which closes a switch to supply a low voltage to the load regulator motor.

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- The load regulator motor moves the variable resistor to increase the main generator field strength and therefore its output.
- The load on the engine increases so its speed falls and the governor detects the reduced speed.
- The governor weights drop and cause the fuel rack servo system to actuate.
- The fuel rack moves to increase the fuel supplied to the injectors and therefore the power from the engine.
- The lever (mentioned above) is used to reduce the pressure of the governor spring.
- When the engine has responded to the new control and governor settings, it and the generator will be producing more power.
- On locomotives with an alternator, the load regulation is done electronically.
- Engine speed is measured like modern speedometers, by counting the frequency of the gear teeth driven by the engine, in this case, the starter motor gearwheel.
- Electrical control of the fuel injection is another improvement now adopted for modern engines.
- Overheating can be controlled by electronic monitoring of coolant temperature and regulating the engine power accordingly.
- Oil pressure can be monitored and used to regulate the engine power in a similar way.

Diesel Loco Shed, Vijayawada was established in the year 1979 at the cost of about Rs.170 lakhs to maintain 20 WDS4 locomotives and increased to 35 in the year 1989. Progressively these locomotives were withdrawn as WDM2 de-rated locomotives were used for shunting purposes. Meanwhile 02 locos in the year 2000 and 15 in the year 2004 have been transferred to Western Railway. Again 02 locos of Diesel Loco Shed, Moulali were transferred to this shed in the year 2004 leaving the effective holding of WDS4 locomotives at BZA to 20. In the year 2006, 13 locomotives have been condemned and auctioned and 02 locomotives were transferred to Shakurbasti Shed, Northern Railway. Presently 05 WDS4 locomotives are under premature condemnation. Railway Board had sanctioned vide letter No.M.66/DSL/COND.underaged/ 120/Dec.06 dt.24.02.10. With the reduction of WDS4 locomotives and increase in Electric traction in BZA Division, DEMU services started in Vijayawada from the year 1996 with 5 consists placed under Diesel Loco Shed, Vijayawada to ease severe traffic constraints both in passenger/express services and also in suburban traffic. Presently the total

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holding of DEMUs is 31 DPCs, 13 DTCs and 54 TCs and is working in BZA, GNT, GTL and SC divisions.

In the year 2002 apart from DEMUs and WDS4 locomotives, 15 WDP1 locomotives were transferred from Diesel Loco Shed; Kazipet to this shed which are working in Mail/Express/Passenger services in BZA, GNT and GTL divisions. During the month of October 2008, 02 more WDP1 locomotives were transferred from Diesel Loco Shed, Tuglakhabad, and Northern Railway and put in BZA loco links from 01.11.08 & 09.11.08.

In March 2007, 2 RAIL BUSES were received from NED division to run between COA-KLPH sections of BZA division.

In addition to these locos, 12 WDM2 over aged locomotives were transferred from DLS/KZJ, DLS/GTL for utilizing in Coaching shunting, Yard Shunting, and BT etc. over BZA & GNT division with effect from 01.12.2009 and also 04 WDM2 over aged locomotives were transferred from DLS/GTL with effect from 01.02.10.

a) Year of commencement:

07.08.1983

b) Road No./Type of the first loco homed in shed:

19524/WDS4

c) Type-wise holding

WDP1	17
WDM2 (Main Line)	3
WDM2 (over-aged)	12
DPCs	31
Rail Buses	2
SPART	1
140T BD Crane	1
DEMU coaches (DPC, DTC & TC)	93

Table 2: Details of Locomotives and Rolling Stock at South Central Railway

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d)Maximum Holding (Year/Number of Locos)

Type of Stock	Year	received			Cumulative Holding			
WDP1	2002	15			17			
	2008 (From DLS/TKD)	02						
WDM2 (over-aged)	2009 (from DLS/KZJ)	07			12			
	2010 (from DLS/GTL)	05						
		DPCs	DTC	TC	DPC	DTC	TC	
DEMUs	1996	5	5	5	5	5	5	
	1998	11	5	15	16	10	20	
	1999	1	0	0	17	0	20	
	2001	4	1	7	21	11	27	
	2002	3 (HHP)	0	9	24	0	36	
	2003 (Tfd. From S.Rly)	5	2	9	29	13	45	
	2004	2 (Aerodynamic)	0	6	31	-	51	
	2006	0	0	1	-	-	52	
	2008	0	0	2	-	-	54	
	2012	-2(Aerodynamic)	0	-6	29	-	48	
	2012	2(A/D with CDTS)	0	6	31	-	54	
		Total	31	13	54	98		
	Rail Bus	2007	02			02		
SPART	2001	01			01			
140T BD crane	1990	01			01			

Table 3 : Year Wise Distribution of DEMU Units and Other Rolling Stock

e) Present Loco link

i. WDP₁ LOCOMOTIVES:

Sl. No	No. of Locos	Link No	Sections covered
1	09 WDP ₁ locomotives	1	BZA-MTM-NS-RJY-NDKD-RAL-DKD-MCLA-GNT.
2	05 WDP ₁ locomotives	2	BZA-MTM-BVRM-CCT

i. WRB₁ (RAIL BUS) (Commissioned on 18.05.07)

Sl. No	No. of Locos	Link No	Sections covered
1	01 Rail Bus	RC 1 & 2	COA-KLPH

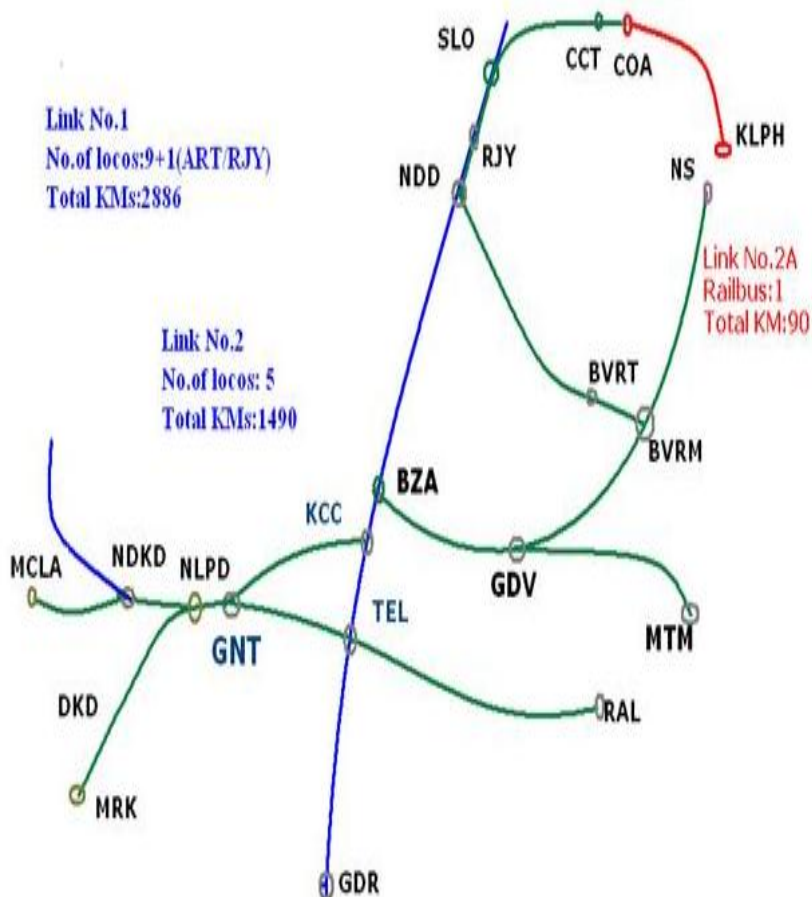


Fig19 : WDP1 Locomotive and WRB1 Rail Bus Operation Links and Sections Covered

ROUTE MAP OF WDP1 LINK HOMING DIESEL SHED-BZA

ii. WDM2 (over-aged) locos w.e.f.01.12.09/01.02.10:

lised in BZA & GNT division for shunting and BT purpose

iii. DEMU:

Sl. No	Link No	as per Link Chart			Actual			Sections covered
		No. of Units	No. of Rakes	No. of Power Cars	No. of Units	No. of Rakes	No. of Power Cars	
1	247	3	4	12	3	4	12	BZA-BVRM-NS; NS-NDD-RJY ;
2	247A	3	1	3	3	1	3	BZA-GDV- BZA-MTM-BZA
3	250	2	2	4	2	2	4	BZA-GNT-TEL-RAL;
4	4	2	2	4	2	2	4	TPTY- KPD-V.Cont.
5	6	2	2	4	2	2	4	BZA-KMNR- JGTL-PDPL-SRUR
Total		12	11	27	12	11	27	

Table 4: Summary of Rake Composition and Section Coverage -As per Link Chart and Actual Data

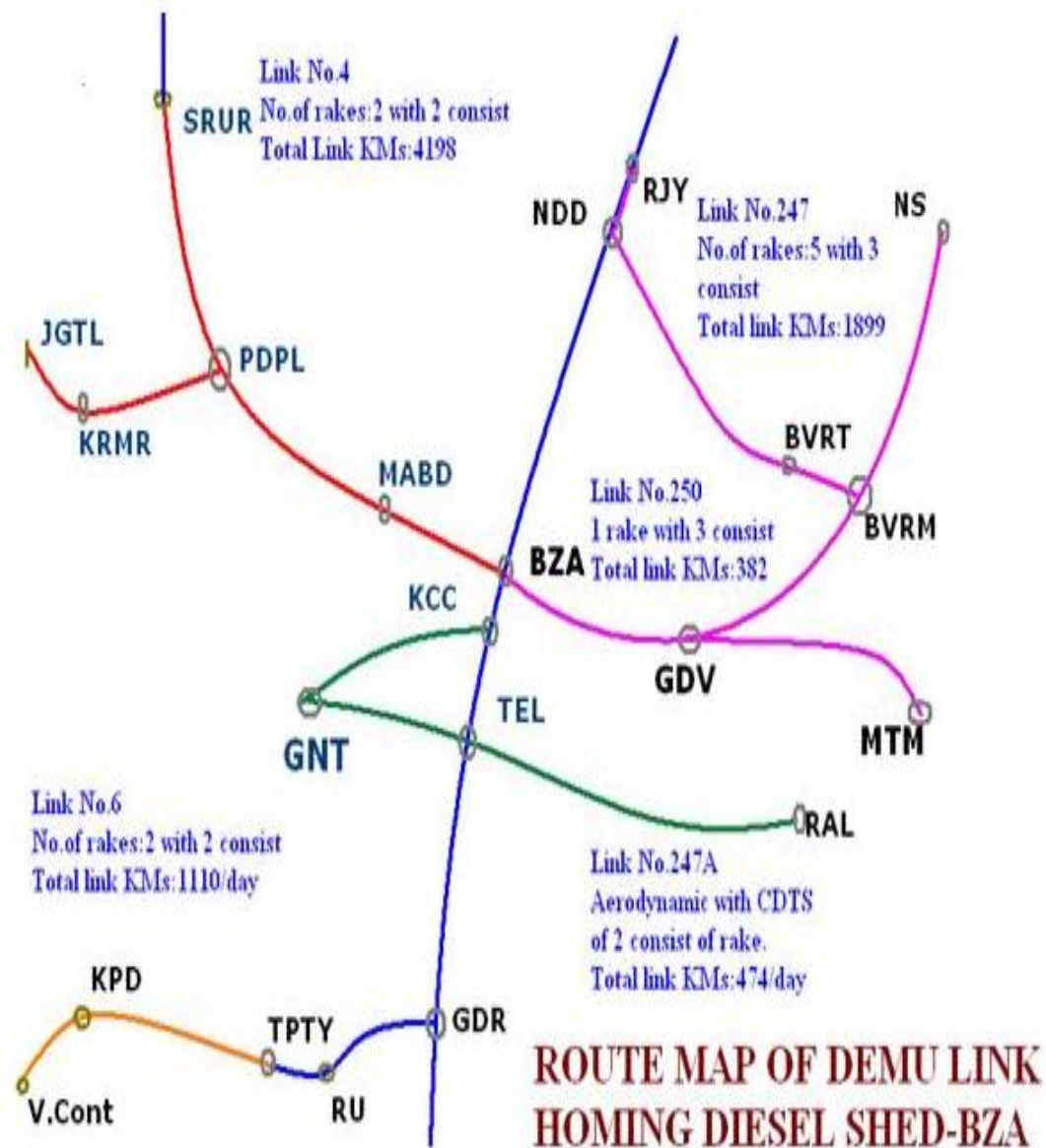


Fig20 : Route Map of DEMU Links under Homing Diesel Shed

5. Homing Capacity:

Originally planned for homing 35 WDS4 locomotives, subsequently one 198 meters pit line was constructed to place 9 coach rake formation of DEMU in the year 2000.

6. Augmentation Plans:

One more 198 meters pit line is under construction for DEMU rake schedules under PWP No. 298(LAW) 08-09.

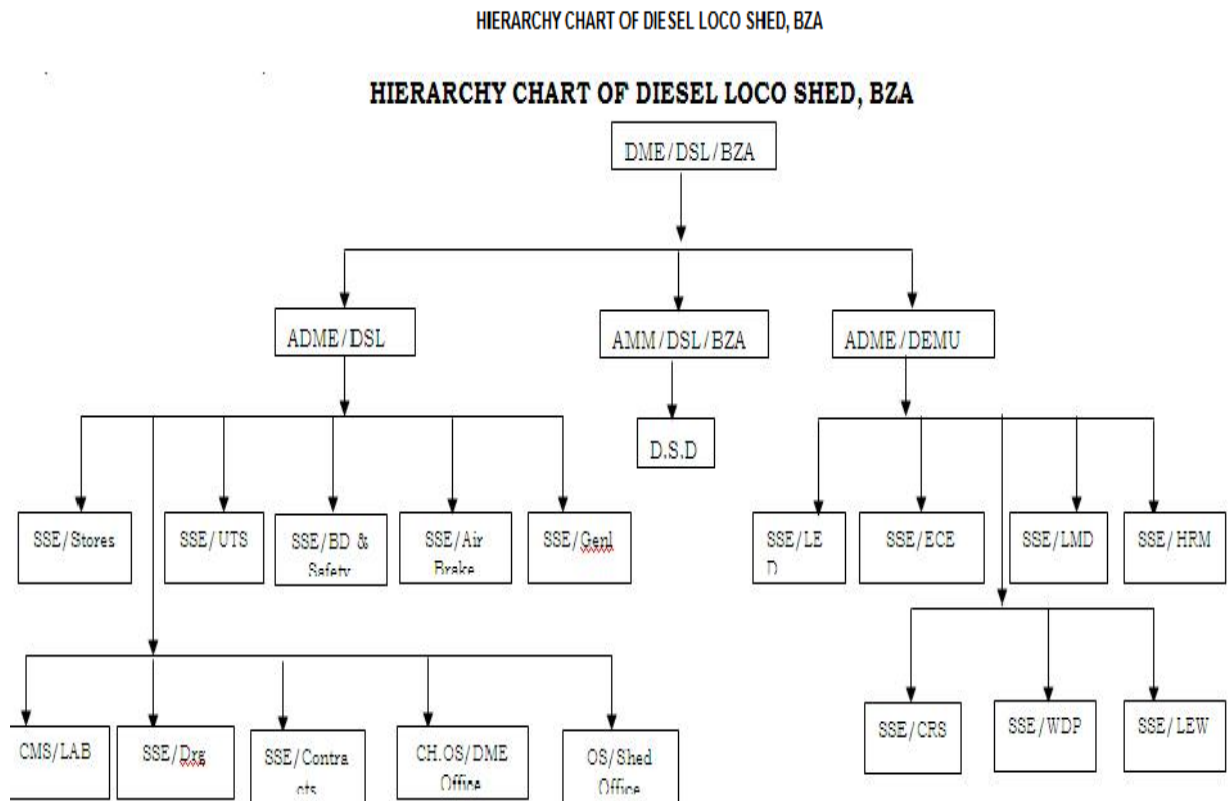


Fig21 : Organization Hierarchy Chart of Diesel Loco Shed

7. Centre for training to Loco Pilots on DEMUs:

Diesel Loco Shed, Vijayawada is imparting trainings of initial/Refresher course to Loco Inspectors, Loco Pilots/Asst. Loco Pilots of other Divisions/Railways on DEMUs regularly besides BZA division Running staff/Loco Inspectors.

8. Performance Parameters

Specific fuel consumption::

WDP1	DEMU	WDM2(over-aged)	
		Goods	Chg.
3.83	2.05	1.81	1.51

Lube oil consumption::

WDP1	DEMU	WDM2(over-aged)
0.69	0.48	0.83

9. **Shed consumption of fuel**: 7525 liters/month

10. **KmS. Earned by Shed Locos/month**:

WDP1	DEMU
150624	315470

7 IOH (COACH CARE CENTER)

AIR BRAKE SYTEM:

INTRODUCTION

- In Air Brake system compressed air is used for operating the brake system.
- The locomotive compressor charges the feed pipe and the brake pipes throughout the length of the train.
- The feed pipe is connected to the auxiliary reservoir and the brake pipe is connected to the brake cylinder through the distributor valve.
- Brake application takes place by dropping the pressure in the brake pipe.

7.1 PRINCIPLE OF OPERATION OF TWIN PIPE GRADUATED RELEASE AIR BRAKE SYSTEM

Charging the brake system

- _ Brake pipe through the length of train is charged with compressed air at 5 Kg/cm².
- _ Feed pipe throughout the length of train is charged with compressed air at 6 Kg/cm².
- _ Control reservoir is charged to 5 Kg/cm².
- _ Auxiliary reservoir is charged to 6 Kg/cm².

Brake application stage

For brake application the brake pipe pressure is dropped by venting air from the driver's brake valve. Subsequently the following actions take place.

- The control reservoir is disconnected from the brake pipe.
- The distributor valve connects the auxiliary reservoir to the brake cylinder and the brake cylinder piston is pushed outwards for application of brakes.
- The auxiliary reservoir is continuously charged from feed pipe at 6 Kg/cm².

Description Reduction in B. P. Pressure

Minimum Brake Application 0.5 to 0.8Kg/cm²

Service Brake Application 0.8 to 1.0Kg/cm²

Full Service Brake Application 1.0 to 1.5Kg/cm²

Emergency Brake Application Brake pipe is fully exhausted and its pressure reduces to almost zero.

Brake release stage:

- Brakes are released by recharging brake pipe to 5 Kg/cm² pressure through the driver's brake valve.
- The distributor valve isolates the brake cylinder from the auxiliary reservoirs.
- The brake cylinder pressure is vented to atmosphere through DV and the Brake cylinder piston moves inwards.

AIR BRAKE SUB ASSEMBLIES

The various Air Brake sub-assemblies and components are:

- i) Brake pipe and feed pipe
- ii) Cut-off angle cock
- iii) Brake cylinder
- iv) Dirt collector
- v) Auxiliary reservoir
- vi) Distributor valve
- vii) Isolating cock
- viii) Check valve

AIR BRAKE HOSES (Brake Pipe & Feed Pipe Hoses):

- To maintain continuity throughout the length of train, the brake pipe (BP) and feed pipe (FP) are fitted with flexible hoses.
- For easy identification, coupling heads are painted with green colour for B.P and white colour for F.P.

CUT OFF ANGLE COCK :

- Cut off angle cocks are provided both on brake pipe & feed pipe on either end of each coach to facilitate coupling and uncoupling of air hoses.
- When the handle of the cut off angle cock is placed in closed position it cuts off the passage of compressed air, thereby facilitating coupling and uncoupling action.

BRAKE CYLINDER:

- On every coach fitted with air brake system two brake cylinders are provided for actuating the brake rigging for the application and release of brakes.
- During application of brakes the brake cylinder develops mechanical brake power by outward movement of its piston assembly after receiving air pressure from Auxiliary reservoir through the distributor valve. This is transmitted to the brake shoes through a combination of levers.

- During release action of brakes the compression spring provided in the brake cylinder brings back the rigging to its original position.

AUXILIARY RESERVOIR:

- On both the ends of the reservoir, flanges are provided for pipe connections. One end of the auxiliary reservoir is connected to the brake pipe through the distributor valve.
- Auxiliary reservoir is charged through the feed pipe to a pressure of 6kg/sq cm. At the bottom of the auxiliary reservoir, a drain cock is provided for draining out the condensate /moisture.

DIRT COLLECTOR

- Dirt Collector is placed in the brake pipe line and feed pipe line at a point from where a branch is taken off to the distributor valve and the auxiliary reservoir.
- The air entering into the dirt collector from the brake pipe and feed pipe is guided through suitably shaped passages in the dirt collector body to produce centrifugal flow.
- The air is then filtered through additional filter assembly before it is passed to outlet on branch pipe side to provide dust proof air to the distributor valve auxiliary reservoir after arresting fine dust particles.
- The dirt contained in the air descends down and gets deposited in the dirt chamber.

DISTRIBUTOR VALVE:

- Distributor valve is the most important functional component of the air brake system and is also sometimes referred to as the heart of the air brake system.
- The distributor valve senses drop and rise in brake pipe pressure for brake application and release respectively.
- It is connected to the brake pipe through branch pipe.
- Various other components connected to the distributor valve are auxiliary reservoir, brake cylinders and control reservoir.

Function of Distributor Valve:

For application and release of brakes the brake pipe pressure has to be reduced and increased respectively with the help of driver's brake valve. During these operations the distributor valve mainly performs the following functions.

- Charges the air brake system to regime pressure during normal running condition.
- Helps in graduated brake application, when pressure in brake pipe is reduced in steps.
- Helps in graduated brake release, when pressure in brake pipe is increased in steps.

- Quickly propagates reduction of pressure in brake pipe throughout the length of the train by arranging additional air pressure reduction locally inside the distributor valve.
- Limits maximum brake cylinder pressure for full service application/ emergency application.
- Controls the time for brake application and brake release depending on service conditions
- Facilitates complete discharge of air from the air brake system manually with the help of operating lever.
- Protects overcharging of control reservoir when the brake pipe pressure is quickly increased for releasing the brakes.

Three designs of distributor valves are in use on coaches. These are:

- C3W Type distributor valve
- KE type distributor valve.
- P4aG type distributor valve.

*In locos C3 distributor valve is used:

C3W DISTRIBUTOR VALVE:

Operation of C3w Distributor Valve:

For effective functioning of the air brake system, the distributor valve has to operate effectively during:

1. Charging stage
2. Application stage
3. Release stage

Charging Stage

- During charging stage the compressed air flows from the brake pipe and enters into the brake pipe chamber of the main valve, cut off valve and quick service valve.
- Due to this pressure the various valves get, activated and perform their functions.

Application Stage

EMERGENCY APPLICATION

- During emergency application the brake pipe pressure is reduced rapidly to 0 kg/cm² by the driver's brake valve.
- Because of this drop the position of the various valves will be altered.

GRADUATED APPLICATION

- During graduated brake application the brake pipe pressure is dropped in steps by driver's brake valve.
- The movement of various valve assemblies is almost in the same direction as during emergency application, but their movement is comparatively less. Release Stage
- When the brake pipe pressure is increased in steps for graduated release of brakes the position of the different valves is altered.

Manual release:

- Double release valve provides for accelerated manual brake release, which is particularly useful during shunting operation.
- A short pull on the lever of double release valve is all that is needed.

PASSENGER EMERGENCY ALARM SYSTEM

Passenger emergency alarm system consists of two components:

- (i) Passenger Emergency Alarm Signal Device (PEASD).
- (ii) Passenger Emergency Alarm Valve (PEAV).

These two components in combination give an indication to the driver that some passenger is in need to stop the train. The indication is transmitted from the coach when the passenger pulls the chain.

CHECK VALVE:

- Check valve with choke (non-return valve) is fitted in the branch line of feed pipe before auxiliary reservoir.
- Check valve allows flow of air in one direction as indicated by the arrow on the body and reverse flow of air is prevented thus avoiding fall in auxiliary reservoir pressure.

ISOLATING COCK:

- Ball type isolating cocks are provided on the feed pipe branch line before the auxiliary reservoir as well as on the branch pipe leading to brake cylinders and passenger emergency alarm valve.
- Isolating cocks can be operated to cut-off air supply to these equipments as per the service requirements.

CONCLUSION:

By this summer industrial training in south central railways we have come to know about the functioning of railways in Vijayawada. We have observed and learned many aspects in safety measures, signal & communication system, maintenance and service system of coaches and trains, loco pilot training and functions, train functioning and tracking systems, electrical traction system, sub stations of railways in Vijayawada division. By observing and analyzing these systems, they are very helpful for us in developing new technological systems & upgrading the technological equipments with present equipment for better functioning of railways & better service to people.



SOUTH CENTRAL RAILWAY
WOMEN'S WELFARE ORGANISATION [REGD]
VIJAYAWADA



REGD No: 183/1992

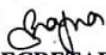
DATE: 07-07-2025

TO WHOM SOEVER IT MAY CONCERN

This is to certify that Mr. VENDRA LAKSHMI NARASIMHA
S/o Sri. VENDRA VENKATA SRINIVAS Registration No: 23K65A0224 course
BACHELOR OF TECHNOLOGY – EEE – semester IV-1, student of SASI
INSTITUTE OF TECHNOLOGY & ENGINEERING (AUTONOMOUS),
TADEPALLIGUDEM has successfully completed the INTERNSHIP in
ELECTRICAL DEPARTMENT, SOUTH CENTRAL RAILWAY,
VIJAYAWADA from 12TH MAY 2025 to 05TH JULY 2025.

During the tenure of his internship, his conduct and contribution
have been GOOD.

We wish him all the best and success.


SECRETARY
SOUTH CENTRAL RAILWAY
WOMEN'S WELFARE ORGANISATION
VIJAYAWADA


PRESIDENT
SOUTH CENTRAL RAILWAY
WOMEN'S WELFARE ORGANISATION
VIJAYAWADA

